

IAL Biology Unit 5 (WBI15/01) — Complete A* Study Note

Respiration, Internal Environment, Coordination and Gene Technology

Built entirely from analysis of 20 past papers and their mark schemes, June 2019 → June 2026.

Everything below is traceable to what examiners actually wrote. Where a mark scheme states an **ACCEPT**, **IGNORE**, **REJECT** or **DO NOT ACCEPT** ruling, it is quoted — those rulings are where marks are won and lost. Question references are given as *series + question number* (e.g. *Jan 2023 Q3(c)*) so you can go straight back to the paper.

The 20 papers are: **June 2019 and Oct 2019** (legacy WBI05 "Energy, Exercise and Coordination", 30-mark article question), and **18 papers on the current WBI15 specification**, Oct 2020 → June 2026. Where the legacy papers test the same content, they are used; where they test old-spec-only content, they are left out.

Contents

§0 The paper, and what the mark schemes reward

Part A — Respiration (§1–§8)

Part B — Muscles, heart, ventilation and exercise (§9–§17)

Part C — Homeostasis and the kidney (§18–§25)

Part D — Nerves, synapses, brain, eye and behaviour (§26–§36)

Part E — Plant responses (§37–§39)

Part F — Gene expression and gene technology (§40–§46)

Part G — The scientific article question (§47–§48)

Part H — Practical and data skills (§49–§51)

Part I — Exam technique: levels questions, command words, calculations, phrase bank, the 25 most lost marks, per-paper map, A* plan (§52–§58)

0. THE PAPER, AND WHAT THE MARK SCHEMES REWARD

Code	WBI15/01
Time	1 hour 45 minutes
Marks	90

Structure	8 questions (9 in Jan 2026). Questions 1–7 are worth 70 marks : multiple-choice, short-open, calculations, and one 6-mark levels-based (*) question . The last question is worth 20 marks and is set on the pre-released scientific article
Legacy WBI05 (2019)	7 questions; article question worth 30 marks
Asterisked (*) questions	One per paper on WBI15, always 6 marks, marked by levels , not points
Maths	Every current paper carries roughly 8–12 marks of calculation (percentage change, magnification, ratio, cardiac output, RQ, area/volume)
Synoptic	The article question freely draws on Units 1–4 (immunity, protein structure, natural selection, inflammation, cell structure)
Equipment	Scientific article insert, calculator, ruler

Pace: 90 marks in 105 minutes $\approx$ **1.17 min per mark**. Budget **about 25 minutes for the 20-mark article question** and do not let Questions 1–7 eat into it.

Topic weighting, counted across the 18 WBI15 papers

Every sub-question of every paper was tagged by topic and its marks summed. The article question (20 marks) is counted separately, because it can test anything.

Topic (Questions 1–7 only)	Mean marks/paper	Range	Appears in
Neurones, action potentials, synapses, drugs	10	6–20	17 of 18
Heart, cardiac output, control of heart rate and ventilation, exercise	10	2–15	Every paper
Kidney, ADH and osmoregulation	8	4–17	14 of 18
Respiration, chemiosmosis, lactate, RQ	8	4–14	17 of 18
Gene expression, epigenetics, gene technology, GM	7	3–21	15 of 18
Muscle structure, sliding filament, fibre types, joints	7	3–14	14 of 18
Plant responses: phytochrome, auxin, gibberellin	4–5	1–13	11 of 18
Brain regions, brain imaging, Parkinson's/dopamine	4	1–12	11 of 18
Eye: rods, rhodopsin, pupil reflex	4	3–9	11 of 18

Topic (Questions 1–7 only)	Mean marks/paper	Range	Appears in
Homeostasis, thermoregulation, hormone action	4	2–13	11 of 18
Habituation and learning	1–2	1–7	5 of 18
Scientific article question	20	fixed	Every paper

Read that table as a revision order: **nerves, heart, kidney and respiration** carry about 36 of the 70 non-article marks in a typical paper.

The five things the Unit 5 mark schemes reward that students miss

1. **Exact wording.** The Unit 5 schemes reject everyday words again and again: "*do not accept signals / messages*" (for impulses), "*REJECT produce energy*" (energy is released), "*do not accept sodium unqualified*" (write sodium **ions**), "*REJECT ATPase*" (write ATP synthase), "*IGNORE control centre unqualified*" (name the centre).
2. **Chains in the right order.** Several schemes state "*mps need to be in logical order to gain credit*" (sliding filament, Oct 2022 Q2(b)(ii), Oct 2023 Q7(a)(ii)). A correct fact in the wrong place in the sequence does not score.
3. **Statistics named, not waved at.** Almost every "comment on" question has a mark for error bars or SD — but "*don't accept {overlapping / not overlapping} SDs are (not significant / significant) unqualified*" (June 2023 Q3(b)(i)). You must say **which** groups, and **what** the overlap means.
4. **A methodology comment.** From 2023 onwards, "comment on" and levels questions routinely credit a point such as "*no indication of sample size / age / sex*" or "*no error bars so cannot comment on significance*". It is often the easiest mark on the question.
5. **Explanation below the level of the description.** Levels-based schemes say it outright: "*Level 3: detailed mechanism ... at a sub-cellular level eg. effect on transcription*" (June 2021 Q7(b)), "*at molecular level*" (Jan 2022 Q5(b)). Describing the graph gets Level 1; explaining the physiology gets Level 2; the molecular mechanism gets Level 3.

PART A — RESPIRATION

1. ATP — what it is and how it releases energy

MCQ fact	Mark-scheme wording
Energy is released by hydrolysis of ATP	<i>June 2019 Q5(a)(i), June 2021 Q1(c)(i): "C is not correct because phosphorylation of ADP requires energy"</i>
ATP is made by phosphorylation of ADP	<i>June 2025 Q2(a)(ii): the inner mitochondrial membrane's role is "to phosphorylate ADP"</i>
Myosin contains the enzyme that hydrolyses ATP	<i>Jan 2021 Q4(a)(iv)</i>
Enzymes are proteins	<i>June 2023 Q2(a)(ii)</i>

"Describe how ATP is used to supply energy" (Jan 2022 Q1(c), 2 marks):

hydrolysis of ATP / of the phosphate bond · **releases energy**

High-energy phosphates (June 2021 Q8(e)): ATP and **creatine phosphate** — "*creatine phosphate provides phosphate to ADP to produce more ATP to continue contraction.*"

2. Glycolysis — in the cytoplasm

1. Glucose is **phosphorylated** using **2 ATP** (phosphorylated hexose) — *Oct 2020 Q1(a)(i):* "phosphorylation of hexoses" is the step in anaerobic respiration.
2. The hexose phosphate splits into **two triose phosphate** (3C) molecules.
3. Triose phosphate is **oxidised** — hydrogen removed and accepted by **NAD**, forming **reduced NAD**.
4. **4 ATP** are made by **substrate-level phosphorylation** → **net 2 ATP**.
5. End product: **2 pyruvate** (3C).

The diagram MCQ (Oct 2025 Q1(a)(ii)): glucose → phosphorylated hexose → pyruvate, with W → X and Y → Z. Answer **A: W = ADP, X = ATP, Y = NAD, Z = reduced NAD** (in the pyruvate-forming step ADP is phosphorylated and NAD is reduced).

NAD in both stages (June 2019 Q5(a)(ii)): "*NAD accepts electrons in glycolysis and in the Krebs cycle*" — "*B is not correct because it is NAD not reduced NAD that accepts electrons.*"

3. The link reaction and Krebs cycle — in the mitochondrial matrix

3.1 Link reaction — the chain the mark scheme wants (Oct 2025 Q1(b)(i), 3 marks)

pyruvate **moves into the matrix** · pyruvate is **oxidised by NAD** — "*ACCEPT dehydrogenated / NAD is reduced to NADH / H atoms are removed*" · pyruvate is **decarboxylated** — "*ACCEPT carbon dioxide is removed*" · the **acetyl group combines with coenzyme A** to form **acetyl coenzyme A** · which **enters the Krebs cycle**

- The conversion of pyruvate to a 2-carbon compound requires **oxidised NAD** (Jan 2025 Q7(b)(iii)) — not reduced NAD, not NADP.
- Respiratory substrates enter Krebs "**as molecules containing 2 carbon atoms produced by the link reaction**" (Oct 2020 Q1(b)(i)).
- **Location MCQ (Oct 2021 Q5(a)(i))**: glycolysis — **cytoplasm**; link reaction — **matrix**; Krebs — **matrix**.
- **Order (Jan 2024 Q1(a))**: glycolysis → link reaction → Krebs cycle.

3.2 Krebs cycle

Acetyl (2C) + **oxaloacetate** (4C) → citrate (6C) → back to oxaloacetate, releasing **2 CO₂** by decarboxylation, producing **reduced NAD**, **reduced FAD** and **one ATP** (substrate-level) per turn.

"Name three products of Krebs cycle" (Jan 2025 Q7(c), 2 marks — "**any three = 2 marks**"): carbon dioxide · ATP · reduced NAD · reduced FAD · oxaloacetate.

"Describe the role of the Krebs cycle in the production of ATP" (June 2019 Q5(b)(iii), 3 marks):

Krebs cycle **produces reduced NAD** (and reduced FAD) · **electrons are transferred to the electron transport chain** · reference to **oxidative phosphorylation** · some ATP is made by **substrate-level phosphorylation** — "*ALLOW some ATP is also produced directly*"

Stage products table (June 2021 Q1(a), 3 marks): glycolysis — **pyruvate / reduced NAD**; Krebs — **reduced NAD / reduced FAD / CO₂**; oxidative phosphorylation — **water / NAD / FAD**.

4. Oxidative phosphorylation and chemiosmosis — the most-asked chain in Unit 5

Asked as a full chain in Oct 2019 Q3(b) (5), Oct 2020 Q1(b)(iii) (5), Jan 2022 Q1(b) (3), Oct 2022 Q1(b) (2), Jan 2023 Q8(b) (3), June 2023 Q2(a)(iii) (2), June 2025 Q2(b) (3) and June 2026 Q8(d) (3). If you learn one chain word-perfect, learn this one.

4.1 The chain examiners want

1. **Reduced NAD and reduced FAD** carry **hydrogen** to the **electron transport chain** on the **inner membrane (cristae)**.
2. The hydrogen splits into **protons (H⁺) and electrons** — Jan 2023 Q8(b): "hydrogen (atoms from reduced co-enzymes) split into {H⁺ / protons} and electrons".
3. **Electrons pass along the electron transport chain, releasing energy** — June 2025: "REJECT produce energy".
4. The energy is used to **pump protons from the matrix into the intermembrane space**, building a **proton (electrochemical) gradient**.
5. Protons **diffuse back into the matrix through ATP synthase** — chemiosmosis.
6. ATP synthase catalyses **ADP + Pi → ATP** (phosphorylation of ADP).
7. **Oxygen is the final electron acceptor**; it combines with electrons and protons to form **water**.

Rulings that decide the marks:

- "IGNORE reference to $ADP + Pi \rightarrow ATP$ without reference to ATP synthase" (June 2025 Q2(b), June 2026 Q8(d)). The ATP-making mark needs the enzyme named.
- "REJECT ATPase" (June 2026 Q8(d)). Older schemes accepted "ATP synthetase / ATP-ase" (Oct 2019) or ignored it (Oct 2020); the current scheme rejects it. **Always write ATP synthase.**
- "ACCEPT actively transport protons for pump protons"; "ACCEPT H⁺ / hydrogen ions" for protons — but "IGNORE hydrogen / H₂ / H" when the box asks for protons (Oct 2019 Q3(a)(ii)).
- Location of the ETC: **inner membrane / cristae** — "IGNORE membrane or intermembrane space" (Oct 2019 Q3(a)(i)).

4.2 The chemiosmosis MCQ (Jan 2025 Q7(b)(i))

Correct: **"hydrogen ions diffuse from the intermembrane space to the matrix through ATP synthase."** Distractors: ions *pumped* from matrix to intermembrane space *through ATP synthase* (wrong — pumping is by the carriers); "to the **stroma**" (that is a chloroplast); ions "jump from inner to outer membrane".

4.3 The role of oxygen — five phrasings, one answer

Paper	Question	Mark-scheme answer
June 2021 Q1(d)	How is oxygen involved in ATP production on the cristae?	terminal electron acceptor · forms H₂O / accepts H ⁺ — "ACCEPT electrons cannot be passed along ETC if no O ₂ to accept them"
Jan 2023 Q5(a)	How is water formed?	reduction of oxygen

Paper	Question	Mark-scheme answer
Oct 2022 Q1(c)	Radioactive O ₂ supplied — first labelled substance?	water (not CO ₂ — the oxygen in CO ₂ comes from the substrate)
June 2025 Q2(c)(i)	Role of oxygen molecules	to accept electrons
June 2026 Q8(d)	Role of oxygen in releasing "more energy"	final electron acceptor → electrons keep flowing → protons pumped → gradient → ATP synthase

4.4 Role of reduced NAD (June 2023 Q2(a)(iii), June 2025 Q2(b))

carries hydrogen (atoms) / electrons / H⁺ to the ETC · reduced NAD is **oxidised / releases electrons** at the inner membrane · the electrons release energy used for **chemiosmosis / oxidative phosphorylation** — "ignore production of energy"

4.5 Drawing a mitochondrion (Oct 2020 Q1(b)(ii), 2 marks)

double membrane with cristae (1) · **matrix labelled** as the site of the Krebs cycle (1). Structure folded from the inner membrane: **crista(e)** — "REJECT matrix, cisternae" (June 2025 Q2(a)(i)).

Inner-membrane statements (Oct 2022 Q1(a)(i)): it is folded (cristae) ✓, location of ATP synthase ✓, site of the ETC ✓, site of protein synthesis **X** → **3 correct**.

4.6 The ETC diagram fill-in (Oct 2019 Q3(a)(ii), 3 marks)

first box **reduced NAD** → second box **NAD** · middle box **H⁺** · fourth box **oxygen (½O₂)** → fifth box **water**.

5. Poisons, inhibitors and mitochondrial disease

This is a favourite way to test the chain in reverse. The logic is always the same: **block the ETC** → **oxygen not used** → **no proton gradient** → **no oxidative phosphorylation** → **cells depend on glycolysis** → **lactate** → **pH falls**.

Inhibitor	Paper	What the scheme credits
Cyanide — inhibits cytochrome oxidase (last carrier)	June 2024 Q6(b), 4 marks	electrons not passed along ETC / reduced NAD not oxidised / chemiosmosis stops · cells respire anaerobically / only glycolysis · insufficient ATP for contraction of heart / intercostal / diaphragm muscles · pyruvate → lactate · lactate lowers pH · enzymes denature

Inhibitor	Paper	What the scheme credits
Cyanide and anaerobic ATP	June 2023 Q2(b)(ii), 2 marks	anaerobic respiration is only glycolysis (substrate-level phosphorylation) · it happens in the cytoplasm and does not involve the ETC
Carbon monoxide — binds haem	June 2023 Q2(b)(i), 3 marks	less ATP · haemoglobin carries less oxygen · cells have less final electron acceptor — <i>"do not accept anaerobic respiration on its own – it needs linking to ATP"</i>
Hydramethylnon (insecticide)	Oct 2021 Q5(a)(iii)	O ₂ consumption decreases , lactate increases
Amytal	Jan 2025 Q7(b)(ii)	O ₂ used decreases , Krebs activity decreases (reduced NAD accumulates, so no NAD for Krebs)
Pyruvate transport inhibitor	June 2022 Q7(b)(ii)	stops pyruvate entering the matrix → no link reaction → CO ₂ production falls (11.6 vs 62.6 nmol) · adding ADP raises the rate because ADP is no longer limiting ATP synthesis (85.0)
MTT assay — reduced NAD turns yellow MTT purple	Jan 2022 Q8(f), 4 marks	use living cells · reduced NAD made in aerobic respiration · same concentrations of both drugs · time to turn purple / colorimeter · the longer it takes, the more toxic

Mitochondrial disease data (Jan 2021 Q4(b), 6-mark *)

Substrates enter the chain at different points: **pyruvate + malate** → **reduced NAD** → **complex I**; **succinate** → **reduced FAD** → **complex II**; **reduced TMPD** → **cytochrome c**. Normal O₂ consumption with a substrate means everything **downstream** of its entry point works.

Individual	Normal O ₂ use with...	Defect located
1	none	in or after cytochrome c
2	all three	before complex I — the ETC is intact
3	succinate and TMPD	complex I
4	TMPD only	complex I, CoQ or complex III (<i>"ignore complex II"</i>)

The levels grid rewards **(1)** descriptions, **(2)** linking each substrate to its entry point, **(3)** locating each defect.

6. Anaerobic respiration, lactate and the oxygen debt

6.1 The chain

1. Without oxygen, the ETC stops, so **reduced NAD is not oxidised** and **NAD runs out**.
2. **Pyruvate is reduced to lactate** by reduced NAD — "**ACCEPT H atoms added to pyruvate**" (Oct 2025 Q1(b)(ii)).
3. This **regenerates NAD**, so **glycolysis can continue**, making **2 ATP per glucose**.
4. Lactate lowers **pH** — MCQ Oct 2020 Q1(a)(ii): "lactate **increases**, causing a **decrease** in blood pH"; Oct 2022 Q1(a)(ii): "muscle becomes **more acidic**".

6.2 What happens to lactate afterwards — asked five times

Paper	Marks	Credited points
Jan 2023 Q6(d)(iii)	2	transported to the liver · converted to pyruvate / glucose · used in link reaction / Krebs / aerobic respiration — " <i>ignore lactate used in aerobic respiration / link reaction / Krebs cycle</i> "
Jan 2024 Q6(a)(ii)	3	lactate diffuses into the blood · oxidised / converted to pyruvate — " <i>ACCEPT dehydrogenation</i> " · pyruvate used in the link reaction / aerobic respiration · in the liver converted to glucose / glycogen
Oct 2025 Q1(b)(ii)	3	pyruvate reduced to lactate · reduced NAD oxidised · lactate to the liver · oxidised back to pyruvate once exercise ends
June 2024 Q6(a)(i) MCQ	1	lactate + oxidised NAD (W) → pyruvate (Z) + reduced NAD (Y)
Oct 2024 Q3(c)(i)	2	lactate stays roughly level during steady exercise because anaerobic production is balanced by conversion to pyruvate / removal by the liver

Trap: lactate is **never** "used in the Krebs cycle". It must be **converted to pyruvate first**. The scheme ignores the shortcut.

6.3 Why oxygen uptake stays high after exercise (June 2019 Q5(b)(ii), 3 marks)

lactate from anaerobic respiration **accumulates** · oxygen is needed to **convert lactate back to pyruvate / glucose** · so that **glycogen stores are replenished** · oxygen used in **aerobic respiration** / to **oxidise reduced NAD** · other oxygen needs — "*repay oxygen debt / re-oxygenate myoglobin / aerobic respiration of breathing muscles / increased metabolic rate*"

6.4 Alcoholic fermentation (Jan 2026 Q9(c), 4 marks)

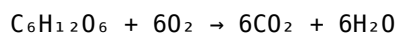
glucose → **pyruvate** (glycolysis) · pyruvate is **decarboxylated** to **ethanal**, which is **reduced to ethanol** · this **oxidises reduced NAD / regenerates NAD** · only **glycolysis** occurs — pyruvate does **not** enter the link reaction or Krebs · overall: glucose → **ethanol + CO₂** — *"Ignore references to lactate"*

7. Energy yield and efficiency calculations

Paper	Data	Working	Answer
June 2021 Q1(c) (ii)	glucose 2867.48 kJ, 38 ATP, 30.51 kJ per ATP	$(38 \times 30.51) \div 2867.48 \times 100$	40.43 %
Jan 2024 Q1(c) (ii)	lipid 12 244 kJ, 122 ATP	$(122 \times 30.51) \div 12\,244 \times 100$, to 3 s.f.	30.4 %

Where does the rest go? (June 2021 Q1(c)(iii)): lost as heat (1) · raising body temperature (1).

Balanced equation for glucose (Jan 2024 Q1(b), 2 marks):



"1 mark if equation correct but not balanced"; "ignore ATP / ADP".

8. Respiratory quotient (RQ)

8.1 Definition — worth a mark on its own

RQ = volume of CO₂ produced ÷ volume of O₂ consumed (in the same time)

- Jan 2026 Q2(a)(i): "carbon dioxide ÷ oxygen (1)" + "**volume** produced and **volume** used (1)" — *"Ignore amount", "Accept moles instead of volume".*
- June 2026 Q5(d)(i): *"IGNORE {number / amount / value} not an acceptable alternative to ratio".*

Substrate	RQ
Carbohydrate (aerobic)	1.0
Protein	about 0.9

Substrate	RQ
Lipid (aerobic)	0.7
Anaerobic respiration of carbohydrate	> 1.0 (CO ₂ produced with little O ₂ taken up)

8.2 RQ calculations

Paper	Working	Answer
Oct 2021 Q5(a)(ii) cockroach	$1.3 \div 1.7 = 0.76$	B, 0.8
Jan 2024 Q1(c)(i) lipid equation $2C_{18}H_{34}O_2 + 51O_2 \rightarrow 36CO_2 + 34H_2O$	$36 \div 51$	0.71
June 2021 Q1(b) balancing a lipid equation	O ₂ and CO ₂ numbers	80 O₂, 57 CO₂ (RQ = $57/80 = 0.71$ — a lipid)
Jan 2023 Q5(b)(i) cyclist, O ₂ uptake 2.20 dm ³ min ⁻¹ , RQ 0.88	2.20×0.88	1.94 dm³ min⁻¹

8.3 Why RQ needs oxygen (Jan 2026 Q2(a)(ii), 2 marks)

CO₂ is produced in the **link reaction and Krebs cycle** · oxygen is needed as the **final electron acceptor / for oxidative phosphorylation** — without O₂ these stages stop, and the equation $CO_2 \div O_2$ cannot be used.

8.4 Interpreting RQ graphs — three versions

- **Cyclist, 90 minutes (Jan 2023 Q5(b)(ii)):** RQ decreased · the cyclist **switched to using lipid / fat** rather than carbohydrate — *"ignore ref to anaerobic respiration"*.
- **Barley grains after germination (Jan 2026 Q2(b)(ii)):** at germination **carbohydrate** · days 0–5 **proportion of lipid increases** · days 5–10 **lipid** is the substrate · after day 10 **carbohydrate increases again**. The question says "use the table" — quote RQ values against substrate proportions.
- **1500 m race (June 2026 Q5(d)(iii), 3 marks):** start RQ $\approx 0.7 \rightarrow$ **lipids** · first ~90–100 s **fewer lipids / more carbohydrate** (aerobic) · after ~100 s RQ $> 1 \rightarrow$ **anaerobic respiration** of carbohydrate for the final sprint — *"ACCEPT because oxygen levels in the muscles have become very low / developed an oxygen debt."* Change in the last 50 s: **0.07**.
- **Carbohydrate–lipid table (Jan 2026 Q2(b)(i)):** as carbohydrate proportion **decreases**, RQ **decreases** — plus a **calculated difference** (e.g. carbohydrate 1.0 $\rightarrow$ 0.0 lowers RQ by 0.3). A relationship without a supporting figure scores only one of the two marks.

8.5 Metabolic rate calculations

- **Field mouse vs elephant seal (Oct 2021 Q5(b)):** mouse $20 \div 10 = 2 \text{ cm}^3 \text{ g}^{-1} \text{ h}^{-1}$; seal $18\,200 \div 5\,000\,000 = 0.00364$; ratio **549–556 times**.
 - **Elephant O₂ per hour (Oct 2022 Q4(a)(ii)):** $1400 \div 5780 = 0.24 \text{ dm}^3 \text{ kg}^{-1}$; $\times 4500 = 1.09 \times 10^3$
— answer **in standard form** for the third mark.
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PART B — MUSCLES, HEART, VENTILATION AND EXERCISE

9. Joints: ligaments, tendons and antagonistic muscles

Structure	Joins	Property	Role, in the scheme's words
Ligament	bone to bone	elastic (contains elastin)	"holds the joint in place" · "maintaining joint stability" · "ACCEPT prevents dislocation" · "elastic (fibres) allow some movement of joint"
Tendon	muscle to bone	inelastic (mostly collagen)	"inelastic so when muscles contract it causes the bones to move" · "ACCEPT inelastic so it transmits force to the bones"
Muscle	attached to bones by tendons	contains actin and myosin	"can contract to move bones" — muscles can only pull
Cartilage	covers bone ends	smooth, shock-absorbing	reduces friction

Sources: Oct 2019 Q4(a) (6 marks: two points for each of muscle, ligament, tendon), Jan 2022 Q2(a), Oct 2022 Q2(a), Jan 2023 Q1, June 2024 Q1(a)(i) (ACL attaches to **bone**).

Statement MCQs that recur:

- Jan 2022 Q2(a)(ii): "antagonistic pairs move the bones" ✓; "muscles acting on a joint are myogenic" ✗ (only cardiac muscle is myogenic); "tendons are more elastic than ligaments" ✗ → **1 correct**.
- June 2023 Q4(a): "tendons act as extensors" ✗; "tendons are myogenic" ✗; "tendons connect muscles to bones" ✓ → **1 correct**.
- Actin and myosin are **proteins** (Oct 2022 Q2(a)(iii)).

9.1 Antagonistic pairs

- The **flexor** (biceps) bends the joint; the **extensor** (triceps) straightens it. The muscle labelled W on the triceps side "acts as an extensor" (Jan 2023 Q1(a)(ii)).
- Straightening the arm (June 2024 Q1(b)(ii)): contraction of the triceps / extensor and relaxation of the biceps / flexor · pulls the bones of the lower arm — "ignore joint" · A and B are antagonistic.**

- **Arm held still (June 2024 Q1(b)(i)):** flexor and extensor both contracted — *"both required for the mark"*.
- **Holding the giraffe's knee steady (Oct 2022 Q2(b)(i)):** muscles work **antagonistically** · both contract (with equal strength) — *"ignore one contracts and the other relaxes"* · ligaments hold the bones in place.

9.2 Collagen and elastin (Oct 2025 Q7(b))

Ligament: 70–80 % collagen, **10–15 % elastin**. Tendon: 75–85 % collagen, **< 3 % elastin**.

both contain a **high proportion of collagen** · collagen provides strength · ligaments contain **more elastin**, allowing stretch / flexibility / some movement of the bones · tendons (almost no elastin) **transmit force** from muscle to bone

Ratio of slopes on the stress–stretch graph (elastin : collagen): $0.45 \div 0.04 = 11.25 : 1$ (accept 11–11.5).

9.3 Joint injury and ageing

- **Damage to a ligament (Jan 2023 Q1(b)(ii)):** a **stated injury** — falling on an outstretched arm, sports tackle, lifting heavy weights, overextending the joint — *"ignore injury unqualified / breaking ligament or bone"*.
- **Reduced tendon stiffness with age (June 2021 Q8(f)):** reduced **contractile force** at the joint · loss of collagen / more elastin · tendon **stretches more** so less movement transmitted · harder to react to avoid a slip or fall.

10. Skeletal muscle structure

A muscle fibre (June 2021 Q8(a), 3 marks): a fibre is a **single cell** · **multinucleate** · **striated** because of repeating bands of **actin and myosin** · made of **myofibrils** divided into **sarcomeres**.

Key parts: **sarcolemma** (cell membrane) · **T-tubules** (carry the action potential into the fibre) · **sarcoplasmic reticulum** (stores Ca^{2+}) · **sarcoplasm** · **many mitochondria**.

The sarcomere is the basic unit of contraction (Jan 2023 Q6(b): "sarcomere", 1 mark).

Band / line	Contains	During contraction
Z line	ends of sarcomere; actin attached	move closer together
I band	actin only	shortens
A band	full length of myosin	stays the same
H zone	myosin only	shortens

Band / line	Contains	During contraction
M line	middle of sarcomere	—

- Jan 2021 Q4(a): structure X (thick filament) is **myosin**; of the four labelled regions only **Z** changes length — the scheme's distractor notes say W, X and Y "do not change in length during contraction".
- June 2024 Q4(a): regions P and R (either side, actin-only) shorten → **C**; structure S (thick filament) is **myosin** — "*ignore tropomyosin*"; ratio relaxed : contracted = **1.8 : 1**.
- June 2026 Q5(a)(ii): sarcomere shorter ✓; H band **wider X** (it narrows); Z lines closer ✓ → **statements 1 and 3**.
- June 2023 Q4(b)(i): one sarcomere is **Z line to Z line** (P to U).

11. The sliding filament theory

11.1 The sequence — in the order the scheme demands

1. An action potential spreads along the sarcolemma and down the **T-tubules**.
2. **Calcium ions are released from the sarcoplasmic reticulum** into the sarcoplasm.
3. **Ca²⁺ binds to troponin**, causing troponin to **change shape**.
4. Troponin **moves tropomyosin, exposing the myosin-binding sites on actin**.
5. **Myosin heads bind to actin**, forming **actin–myosin cross-bridges**.
6. **ADP and Pi are released** from the head → the head tilts: the **power stroke**, pulling **actin towards the M line / centre of the sarcomere**.
7. **ATP binds to the myosin head, breaking the cross-bridge** (head detaches).
8. **ATP is hydrolysed** by the ATPase on the myosin head, and the energy **returns the head to its original position** ("re-cocking").
9. The cycle repeats while Ca²⁺ is present. When stimulation stops, **Ca²⁺ is actively transported back into the SR**, and the muscle relaxes.

Asked as: Oct 2023 Q7(a)(ii) (4 marks — "*mps need to be in logical order to gain credit*"), Oct 2022 Q2(b)(ii) (3), June 2022 Q7(c)(ii) (4), June 2026 Q8(a) (3), Jan 2026 Q8(c)(iv) (3).

The one-line trap, repeated in four schemes: ATP hydrolysis provides energy **to move the myosin head back** — "*Do not accept the power stroke*" (June 2022, Oct 2022, Oct 2023; June 2026: "*IGNORE 'the power stroke' for mp3*"). The power stroke is driven by the **release of ADP and Pi**.

11.2 Role of calcium ions / troponin

- **Calcium ions in contraction (Oct 2021 Q1(a), 2 marks):** Ca²⁺ **binds to troponin** — "**ACCEPT change in shape**" · **tropomyosin moves** · **myosin head can bind** — "**ACCEPT exposing the**

myosin binding site / allowing actomyosin bridges to form".

- **Role of troponin (June 2024 Q4(b), 4 marks):** Ca^{2+} binds to troponin · troponin **changes shape** — "*ignore confirmation change*" · **tropomyosin is moved** — "*ignore released*" · **myosin binding site exposed** · **myosin head binds / cross-bridge forms**.
- **Ca^{2+} binding to troponin C (June 2021 Q8(d)):** tropomyosin changes shape, freeing binding sites · heads bind forming cross-bridges · ATP hydrolysis allows myosin to pull actin inwards.
- **Why a lack of Ca^{2+} causes fatigue (Oct 2024 Q3(a), 3 marks):** less Ca^{2+} binds troponin · troponin doesn't change shape · **tropomyosin stays blocking the binding sites** · **fewer cross-bridges** · **less contraction / sarcomeres shorten less** — "*ignore disrupt contraction*".
- **Statement MCQ (Jan 2021 Q4(a)(iii)):** Ca^{2+} binds troponin ✓ , Ca^{2+} released from SR ✓ , myosin changes shape ✓ → **three**.

11.3 Where the energy goes (Oct 2023 Q7(a)(i))

Energy from ATP is used to **break the actin–myosin cross-bridge** ✓ and to **remove calcium ions from the sarcoplasm** (active transport into the SR) ✓ — **not** to release ADP and Pi.

11.4 Isolated muscle fibres in ATP (Jan 2026 Q8(c))

- Solutions have **the same water potential as the cytoplasm** so there is **no net osmosis** and fibres **do not swell or shrink** — **only ATP / glucose affects length**.
- Control solution: **water and ions / saline** — "*Do not allow glucose / ATP*".
- Percentage decrease from 3 cm to a mean of 2.53 cm: **15.7 %** (range 10–23.3 % accepted from measurement).
- **ATP causes shortening** because ATP binds the myosin head → shape change → head **detaches and re-positions** → binds actin again further along.
- **Glucose shortens fibres less** because it must first be **respired to make ATP** — this **takes time** and yields **less ATP** than supplying ATP directly.

11.5 Force and velocity (June 2023 Q4(c), 3 marks)

as **velocity increases, force decreases** (inverse / negative correlation) · cross-bridges form when **myosin heads bind actin** and are broken when **ATP binds** · in fast contraction **fewer cross-bridges are attached at any one time**, so force is lower

12. Fast-twitch and slow-twitch fibres

Feature	Slow twitch (type I)	Fast twitch (type II)
Respiration	aerobic	anaerobic (glycolysis, creatine phosphate)

Feature	Slow twitch (type I)	Fast twitch (type II)
Mitochondria	many	few
Myoglobin	much — red	little — pale
Capillary network	extensive	sparse
Diameter	smaller	larger
Sarcoplasmic reticulum	less	extensive — more Ca ²⁺ released faster
Glycogen / creatine phosphate	less	more
Contraction	slow, sustained, fatigue-resistant	rapid, powerful, fatigues quickly
Suited to	marathon, posture	sprinting, jumping

The structure questions, as credited:

- June 2023 Q4(b)(iii) (4): fast twitch have **fewer capillaries** — *"ignore less blood supply as not structural"* · **fewer mitochondria** · **less myoglobin** · **larger diameter** / **more SR** · **more glycogen / creatine phosphate**.
- Oct 2025 Q7(a)(i) (3), "adapted for stronger and more rapid contraction": **more glycogen** · **more phosphocreatine** · **extensive SR** · **more Ca²⁺ in the SR** — *"NB fast twitch has more sarcoplasmic reticulum to release calcium ions only scores mp3."*
- Jan 2021 Q2(b)(i) intermediate vs slow: **less myoglobin** · **fewer mitochondria** · **more SR** · **higher glycogen**.
- Jan 2026 Q8(a): "slow twitch store more myoglobin" ✓; "fewer mitochondria" ✗; "fatigue faster" ✗ → **statement 1 only**.
- Jan 2024 Q6(a)(iii): a long-distance runner's fibres have **more mitochondria** (statement 2 only).

Application questions:

- **Marathon runner (Oct 2021 Q1(b), 4 marks):** many mitochondria → **more aerobic respiration / ATP** · myoglobin → **store of oxygen** · larger capillary network → **supply of oxygen and glucose** · slow sustained contraction → **race for longer / less fatigue**.
- **Sprinter (Jan 2022 Q2(b)(ii); June 2026 Q5(c)):** sprinting demands **speed over a short time** · fast twitch **contract more quickly** · rely on **anaerobic respiration** — *"accept contraction does not rely on a supply of oxygen"; "ACCEPT correct refs to glycogen / creatine phosphate"; "IGNORE myoglobin / capillaries"* for the sprinter.
- **Athletes by distance (Jan 2023 Q6(a)):** more slow twitch as distance increases · **more mitochondria** · **larger capillary network** / **more myoglobin** · **aerobic respiration** / **more ATP**.

- **Why numbers of mitochondria differ (June 2022 Q7(c)(i)):** slow twitch have **more** · they rely on **aerobic respiration** · because their **oxygen supply is sufficient** (more capillaries, more myoglobin).
 - **Pronghorn (Oct 2024 Q5(a)(i)):** adaptation for long fast running = **increased slow twitch fibres**.
 - **Training (Jan 2021 Q2(b)(ii)):** training increases **strength and cross-sectional area** · **no effect on slow twitch** · **fast twitch switch to intermediate** · **greater training, greater switch**.
 - **Increased capillary network (Jan 2026 Q8(b)):** more **oxygen and glucose** delivered · glucose = respiratory substrate · oxygen → **aerobic respiration** — "*accept less likely to become anaerobic / reduce build-up of lactic acid*" · **more ATP** · **more CO₂ removed**.
 - **Different myosins from one stem cell (Oct 2025 Q7(a)(ii)):** **differential gene expression** · **transcription factors bind different promoters** · transcription/translation of different myosin genes · **M7 → type I (slow); M2/M1/M4 → type II (fast)**.
-

13. The heart: myogenic contraction and the conduction system

Myogenic means **contraction is initiated within the (cardiac) muscle itself**, without a nerve impulse (*June 2019 Q1(a)(i); June 2021 Q6(a)(ii); Jan 2025 Q6(b)(ii)*: "cardiac muscle that is able to contract with no external stimulus"). The **left ventricle** contracts with the greatest force (*June 2019 Q1(a)(ii)*).

13.1 Coordinated contraction — the 6-mark chain (Oct 2019 Q5(c))

1. Cardiac muscle is **myogenic**.
2. The **sinoatrial node (SAN)** in the right atrium wall is the **pacemaker**; it generates **waves of depolarisation / excitation**.
3. The wave **spreads across both atria**, causing **atrial systole**.
4. A layer of non-conducting tissue stops the wave reaching the ventricles directly; it reaches the **atrioventricular node (AVN)**.
5. The **AVN delays** the wave (so the atria finish emptying).
6. The wave passes down the **bundle of His** in the septum to the **Purkyne fibres**.
7. Ventricles contract **from the apex upwards** — **ventricular systole**.
8. Impulses from the **cardiovascular control centre** regulate the rhythm of the SAN.

Rulings: "**IGNORE impulses**" for the SAN's output — say **waves of depolarisation / excitation** (*June 2019 Q1(c)*: "**IGNORE nerve impulses**"); "**IGNORE SAN is myogenic**" (it is the **muscle** that is myogenic); "**ACCEPT Purkinje**".

Role of the SAN (June 2019 Q1(c), 3 marks): receives impulses from the **cardiac control centre** · acts as the **pacemaker** · produces **waves of excitation** · which **spread across the atria / start atrial systole** · travel to the **AVN**.

Diagram MCQ (June 2021 Q6(a)(i)): Q = SAN, P = AVN.

14. The ECG

Wave	Represents
P	atrial depolarisation → atrial systole
QRS	ventricular depolarisation → ventricular systole
T	ventricular repolarisation → the heart is in diastole (Jan 2022 Q7(b)(i))

14.1 Heart rate from an ECG

heart rate (bpm) = $60 \div \text{time for one cycle (s)}$ — measure R to R, and over several beats if you can.

Paper	Working	Answer
June 2019 Q1(b)(i)	$(11.7 \times 0.2) \div 3 = 0.78 \text{ s per beat} \rightarrow 60 \div 0.78$	76.9 bpm (allow 75.9–77)
June 2021 Q6(b)(iii)	19 beats in 7 s → $60 \div 7 \times 19$	163 bpm — "Do not accept fraction of heart rate"
Jan 2022 Q7(b)(ii)		100 bpm (accept 100–125)
Oct 2023 Q4(b)(i) MCQ		48 bpm
Oct 2025 Q4(a) seal	$(60 \div 3) \times 5 = 100 \text{ bpm before; } 13 \text{ bpm diving}$	change = 87 bpm

- **Very fast heart rate (June 2019 Q1(b)(ii)):** "distance between peaks would be shorter" — "IGNORE more peaks".

14.2 Comparing ECG traces — name the waves

- Oct 2023 Q4(b)(ii): "unless refer to labels PQRST they will not get the marks" — e.g. **smaller P wave in A** · **R peak sharper in A** / **two R peaks in B** · **shorter QRS in A** · **larger T wave in A**.
- June 2021 Q6(b)(iv): "For each mark there must be a clear comparison between the two ECGs" — rhythm · frequency · **size of the R wave** · the **ST segment**.
- Jan 2022 Q7(b)(iii) compare **and** contrast: similarity — **same PQRST shape, R wave of similar height**; difference — patient has **higher heart rate, shorter gap between T and P**.
- Jan 2026 Q6(c): exercise makes the **cardiac cycle shorter, atrial systole shorter, force of ventricular systole greater** — "Ignore references to heart rate / PQRS waves / cardiac output".

- Using an ECG to diagnose POTS (June 2022 Q8(d)): record ECG **before and for 10 minutes after changing posture** · count named peaks per minute · POTS if heart rate rises by 30 bpm / exceeds 120 bpm.
- Showing energy drinks cause irregular rhythm (Jan 2025 Q8(g)): compare **drinkers vs non-drinkers** / before vs after · perform ECG · compare intervals between the same named peaks.

15. Cardiac output

cardiac output = heart rate × stroke volume

Paper	Question	Answer
Oct 2020 Q4(b)	HR difference, swimmers vs control ($4.43 \div 0.0744$ and $4.21 \div 0.0584$)	$72.09 - 59.54 = \mathbf{12.55 \text{ bpm}}$ — " <i>correct number of decimal places and units</i> " for mark 3
June 2021 Q6(b)(ii)	$4500 \div 51$	$\mathbf{88.24 \text{ cm}^3}$
June 2022 Q6(b)(i)	$77 \times 70 = 5390 \text{ cm}^3 \text{ min}^{-1} \rightarrow \times 60 \div 1000$	$\mathbf{320 \text{ dm}^3 \text{ h}^{-1}}$ (2 s.f.)
Jan 2023 Q6(c)(i)	$65 \times 57 = 3705 \text{ cm}^3$	$\mathbf{3.71 \text{ dm}^3 \text{ min}^{-1}}$
June 2025 Q5(b)	$5000 \div 72$	$\mathbf{69 \text{ cm}^3}$ — " <i>69.44 / 69.4 gains 1 mark</i> " (2 s.f. was asked)
June 2026 Q5(b)	$37 \div 128 = 0.289 \text{ dm}^3$	$\mathbf{289 \text{ cm}^3}$
Jan 2026 Q6(a) MCQ		$\mathbf{CO = SV \times HR}$

Cardiac index = cardiac output ÷ body surface area. Units $\text{dm}^3 \text{ min}^{-1} \text{ m}^{-2}$ (June 2022 Q6(b)(ii): individual A $6.7 \div 1.92 = \mathbf{3.5}$; individual B $1.49 \times 2.8 = \mathbf{4.2}$). Jan 2023 Q6(c)(ii): $79 \times 0.075 \div 2.39 = \mathbf{2.48}$; ratio of surface areas $\mathbf{1 : 1.55}$.

Why cardiac output tells you more than heart rate (Oct 2023 Q4(c)(ii)): it includes **stroke volume** — it measures the **volume of blood pumped per minute**; different animals have **different-sized hearts / ventricles**.

16. Control of heart rate — the second most-asked chain

16.1 The chain

1. Exercise → **more (aerobic) respiration** in muscle → **more CO₂** in the blood → **blood pH falls** (also lactate from anaerobic respiration).
2. **Chemoreceptors** (in the **carotid and aortic bodies** and the **medulla**) detect the change. **Baroreceptors** (carotid sinus, aortic arch) detect **blood pressure**.
3. They send **impulses** to the **cardiovascular control centre (CVC) in the medulla oblongata**.
4. The CVC sends **more impulses along sympathetic (accelerator) neurones** to the **SAN**.
5. The SAN **depolarises more frequently** → **heart rate increases**; the **force of contraction** increases → **stroke volume increases**.
6. **Adrenaline** from the adrenal glands also **binds to receptors on the SAN**.
7. To slow the heart: **more impulses via the parasympathetic (vagus) nerve** → SAN depolarises less often.

Asked in: Jan 2021 Q6(d) (slowing after exercise, via **parasympathetic / vagus**), June 2021 Q6(b)(i), Jan 2022 Q3(b)(ii) (whale surfacing), Oct 2022 Q4(b)(i) (cheetah), Jan 2023 Q6(d)(ii), Oct 2023 Q4(c)(iii) (giraffe), June 2023 Q8(b) (vagus), Oct 2024 Q5(a)(ii) (pronghorn, heart **and** breathing), June 2025 Q5(a)(ii), June 2026 Q8(e).

The rulings that cost marks:

- *"Do not accept reference to signals / messages"* — every scheme, often *"REJECT signals / messages once for mp1–3"*.
- *"ignore ref to increased oxygen demand"* as the stimulus (Jan 2023 Q6(d)(ii)) — the receptors detect **CO₂ / pH**, not "demand".
- *"chemoreceptors in medulla oblongata detect increase in blood CO₂ = 2 marks"* (Oct 2022).
- *"ignore muscles unqualified"* — say **SAN depolarises more frequently** or **cardiac muscle contracts more frequently**.
- *"ignore force of contraction"* for breathing muscles (Oct 2024) — the mark is for **more frequent** contraction of intercostals/diaphragm.

16.2 Variants you must be able to adapt

- **Systole and diastole during exercise (June 2025 Q5(a)(ii), 4 marks):** CO₂ ↑ / pH ↓ · chemoreceptors in CVC / medulla / carotid or aortic bodies · more impulses along sympathetic neurones to the SAN (during systole) · **greater force of ventricular contraction** · diastolic pressure stays constant because cardiac muscle is relaxing.
- **Whale surfacing (Jan 2022 Q3(b)(ii)):** a physiological change during the dive — O₂ falls, CO₂/lactate rise, pH falls, oxygen debt · monitored by chemoreceptors · more impulses from medulla to SAN · SAN → higher heart rate.

- **Why diving whales lower heart rate (Jan 2022 Q3(b)(i)):** heart rate **decreases when diving** · the longer the dive, the lower the rate · the gap between surface maximum and dive minimum increases with dive duration.
 - **Drugs after stress in rats (Oct 2021 Q7(c)):** stress raises heart rate · **CVC stimulates the SAN** · both drugs **decrease the frequency of impulses from the SAN** · drug B has the **greater effect**.
 - **Nicotine and adrenaline on Daphnia (Oct 2023 Q6(a)(ii)):** heart rate **decreases as nicotine concentration increases** · **increases as adrenaline concentration increases** · highest rate with **highest adrenaline + lowest nicotine** · *"highest adrenaline concentration and no nicotine caused death of Daphnia"*.
 - **Adrenaline MCQ (Oct 2023 Q6(a)(i)):** breathing rate **increases**, heart rate **increases**.
 - **Why a cheetah can only sprint briefly (Oct 2022 Q4(b)(ii)):** fast twitch fatigue / **cannot get oxygen fast enough** · anaerobic respiration · lactate / oxygen debt builds up (*"accept heat released so body temperature rises rapidly"*).
-

17. Ventilation, exercise physiology and fitness

17.1 Control of ventilation

- **Resting control (June 2019 Q6(b); June 2022 Q8(a)):** stretch receptors in the lungs are stimulated as air enters → impulses **inhibit the inspiratory centre** → **fewer impulses to intercostal muscles / diaphragm** → muscles relax → exhalation. The **ventilation (respiratory) centre in the medulla / pons** sets the basic rhythm.
- **During exercise (June 2023 Q3(c), 4 marks):** more activity / aerobic respiration → **more CO₂ / lower pH** · detected by **chemoreceptors** · **impulses to the respiratory centre / medulla** · **more impulses to intercostal muscles / diaphragm** · **increased ventilation rate** (and depth / tidal volume).
- **Low oxygen (June 2019 Q6(b)):** chemoreceptors detect **low O₂ / low pH / high CO₂** · impulses to the ventilation centre · **more frequent impulses to diaphragm / intercostals** · **increased rate and depth**.

17.2 Ventilation calculations

- **Minute ventilation (June 2023 Q3(b)(ii)):** $0.54 \text{ breaths s}^{-1} \times 60 = 32.4 \text{ min}^{-1}$; $\times 0.040 \text{ dm}^3 = \mathbf{1.3 \text{ dm}^3}$ — *"1.30 = 1 mark only" (2 s.f. asked)*.
- **Alveolar ventilation per hour (Jan 2022 Q7(a)):** $(500 - 150) \times 12 = 4200 \text{ cm}^3 \text{ min}^{-1}$ → $\times 60 = \mathbf{250 \text{ dm}^3 \text{ h}^{-1}}$ — *"max 2 marks if no units / or incorrect units"*.
- **Body mass vs ventilation rate (June 2023 Q3(b)(i)):** **negative correlation** · **small number of species** · **correlation does not imply causation** · a **specific SD comment**, e.g. porcupine (0.27–0.33) and tiger (0.21–0.33) overlap → no significant difference.

- **Lung conditions that change ventilation rate (June 2023 Q3(a)):** asthma, COPD, emphysema, TB, cystic fibrosis, lung cancer, pneumonia, bronchitis.

17.3 COPD and smoking (Jan 2023 Q5(c)(ii), 4 marks)

ventilation rate and heart rate increase · to deliver enough oxygen for **aerobic respiration** · damage to alveolar walls **reduces surface area for gas exchange** · **thick mucus increases diffusion distance** · so less oxygen diffuses into the blood

Calculation: $7.5\% \text{ of } 264\,100 = 1.98 \times 10^4$ people (standard form).

17.4 Blood distribution during exercise

Oct 2019 Q5(b) (5 marks) and Jan 2026 Q6(b)(ii) (4 marks):

cardiac output increases · blood flow to **skeletal and cardiac muscle increases** to supply **oxygen and glucose for respiration** · flow to the **skin increases (vasodilation)** for **heat loss / thermoregulation**, because exercise **produces heat** · flow to the **kidneys and abdominal organs decreases**, so blood can be **redistributed to muscles and skin** · flow to the **brain does not change**, as its oxygen demand does not change

Calculations: % change to abdominal organs ($1200 \rightarrow 800$) = **33 %**; blood in muscles during exercise $5 \text{ dm}^3 \times 68\% = 3.4 \text{ dm}^3$.

Factors to control in an exercise study (Oct 2019 Q5(a)(ii)): length and **type/intensity** of exercise · **fitness / BMI / mass** · **ambient temperature** — *"IGNORE body temperature"* · **food/drug intake** — *"IGNORE gender / age"* (already controlled).

17.5 Training, fitness and recovery

- **Faster recovery in a fit person (Jan 2021 Q6(c)):** greater lung capacity · greater cardiac output · more effective gas exchange and oxygen transport · more rapid repayment of the oxygen debt.
- **Percentage decrease in heart rate (Jan 2021 Q6(a)):** $(170 - 90) \div 170 \times 100 = 47.06\%$; $-16\% = 31\%$ (decimal places appropriate to data).
- **Straight-line extrapolation (Jan 2021 Q6(b)):** $y = (-65 \div 240 \times 300) + 190 = 109 \text{ bpm}$.
- **Resting heart rate falls with training** because **stroke volume increases** (bigger, stronger ventricle) while **resting cardiac output hardly changes** (Oct 2020 Q4(c); June 2021 Q6(b)(ii): $4500 \text{ cm}^3 \text{ min}^{-1}$ before and after, HR $72 \rightarrow 51$).

The four exercise levels questions (Oct 2020 Q4(c); Jan 2024 Q6(b); June 2024 Q6(a)(ii); Jan 2025 Q6(c)) all reward the same explanations. Learn them as a bank:

- athletes have **larger / stronger cardiac muscle** → **greater stroke volume** → lower resting heart rate, higher cardiac output at peak

- athletes have **larger lung capacity / more efficient gas exchange**
- athletes' muscles have **more slow twitch fibres, mitochondria, myoglobin and capillaries** → more **aerobic** respiration at the same power
- non-athletes switch to **anaerobic** respiration sooner → **more lactate / larger oxygen debt** (*June 2024*: non-athlete 10.0 vs athlete 6.4 mmol dm⁻³ at intensity 240)
- a pH fall is detected by **chemoreceptors**, triggering higher ventilation and heart rate
- exercise lowers **blood pressure / cholesterol** → **fewer atheromas** → **lower CVD risk**; improves **insulin sensitivity**; reduces **inflammation** (*Jan 2025*)
- **methodology**: "no error bars / SDs so cannot comment on significance"; "no details about age / gender / health / diet"

17.6 Diving mammals (Oct 2025 Q4(b), 6-mark *)

Seals: arterial and muscle O₂ **fall** during the dive · muscle **lactic acid rises after about 4 minutes** → **anaerobic respiration** · blood lactate **rises after surfacing** as lactate is **flushed from muscle to blood** → **liver** · blood flow to **skeletal muscle falls 99 %**, **cardiac muscle 90 %**, to the **brain rises 20 %** so the **brain keeps respiring aerobically** · **myoglobin stores oxygen** · **heart rate falls** (fewer waves of depolarisation) · aerobic respiration after the dive **repays the oxygen debt**. The levels grid counts **comments, explanations and "general comments"**: 6 marks needs e.g. **3 explanations + 1 general comment**.

PART C — HOMEOSTASIS AND THE KIDNEY

18. Homeostasis and feedback

Homeostasis (Jan 2025 Q5(a), 1 mark): *"maintaining the body in a state of dynamic equilibrium" — "ACCEPT maintenance of a constant internal environment"; "IGNORE example of homeostasis described".*

Negative vs positive feedback (Oct 2021 Q4(a), 2 marks): negative feedback **reverses / decreases** the change from the norm; positive feedback **increases / amplifies** the change.

MCQ (June 2019 Q3(a)): in negative feedback, sensors detect **increases or decreases**, and effectors **reverse** the change.

MCQ (Oct 2021 Q4(b)): at 37 °C *"the body temperature is maintained in a dynamic equilibrium"* — thermoregulation is **continually** required and the body **always** generates heat.

18.1 How negative feedback holds a variable within narrow limits

Jan 2024 Q8(d) (2) and June 2026 Q8(b) (2):

a change is detected by a receptor · impulses to the control centre (hypothalamus / CVC / medulla / pituitary) · an effector counteracts the change, returning the condition to normal · when the receptors detect the condition is back to normal, the effector stops being stimulated

That last point — **switching off** — is the one most answers leave out, and it is what "narrow limits" means.

18.2 Positive feedback in a neurone (Jan 2026 Q3(c), 4 marks)

neurotransmitter binding opens a few sodium ion channels · Na⁺ enters, depolarising the membrane slightly · this opens more (voltage-gated) sodium channels · more Na⁺ enters, further depolarisation · the cycle continues until an action potential / threshold is reached — *"Ignore potassium ions"; "Ignore sodium unqualified – penalise once only"*

18.3 Calcium homeostasis (Oct 2024 Q3(b))

Process: **homeostasis / negative feedback**. Range $\pm 6\%$ of 91 mg dm⁻³ → 85.54 to 96.46 → difference **10.92 mg dm⁻³**.

19. Thermoregulation

19.1 The chain examiners want

1. **Thermoreceptors** (in the skin and in the **hypothalamus**) detect a **change in (core/blood) temperature** from the **set point**.
2. **Impulses** travel to the **thermoregulatory centre in the hypothalamus** (heat-loss or heat-gain centre).
3. The hypothalamus sends **impulses via motor / sympathetic neurones to named effectors**.
4. Effectors **reverse the change**:

Too hot	Too cold
vasodilation of skin arterioles → more blood to skin → more heat lost by radiation	vasoconstriction of skin arterioles
sweating → heat lost by evaporation	sweating stops
hair erector muscles relax	hair erector muscles contract
metabolic rate decreases	shivering ; metabolic rate increases (brown fat)

5. Temperature **returns to the set point** (negative feedback).

Asked in *June 2019 Q3(b)* (exercise), *Oct 2023 Q7(c)* (ostrich), *Jan 2024 Q7(a)* (kangaroo rat 38.5–40 °C), *Oct 2024 Q5(b)(ii)* (human during exercise), *June 2026 Q4(a)(iii)* (emperor penguin, cold).

Rulings: "ignore control centre unqualified" · "ignore signals / messages / information" · "ignore thermoregulators" · "IGNORE location of thermoreceptors" (June 2019) · "IGNORE comments relating to fur / hair" (penguin) · effectors named: "sweat glands, hair erector muscles, liver, smooth muscles in skin blood vessels".

19.2 Variants

- **Seizure raising body temperature (Jan 2021 Q3(c)):** impulses from neurones at the seizure site **reach the thermoregulatory centre in the hypothalamus** · consequence: **reduced sweating / increased metabolic rate / increased respiration**.
- **Rate of heat loss during exercise (Oct 2021 Q1(c)):** exercise **increases the rate of heat loss** · it **levels off / reaches a maximum**.
- **Penguin (June 2026 Q4(a)):** $SA : V = 1.037 \div 0.0706 = 14.69 : 1$ — "If answer given to one decimal place e.g. 14.7 = 0 marks". Other heat-saving features: **dense feathers, fat/blubber, huddling, small extremities** — "IGNORE fur".
- **TRPV1 channels (Oct 2022 Q8(g)):** TRPV1 is **temperature-sensitive** · impulses to the **hypothalamus** · hypothalamus triggers a **named response** (e.g. vasodilation, sweating).

20. Hormones: how they switch genes on

The key distinction, tested in five papers: **can the hormone cross the cell membrane?**

	Peptide hormone (insulin, ADH, AVT, growth hormone, adrenaline)	Steroid hormone (oestrogen, testosterone, oestradiol)
Solubility	water-soluble — cannot cross the membrane	lipid-soluble — diffuses through the membrane
Receptor	on the cell surface membrane	inside the cell — cytoplasm or nucleus
Next step	activates a second messenger (cAMP)	forms a hormone–receptor complex
Gene effect	second messenger activates transcription factors / enzymes	the complex acts as a transcription factor , binding the promoter
Result	transcription → mRNA → translation → protein	RNA polymerase binds → transcription of target genes

20.1 Steroid hormones — mark scheme chains

- **Oestrogen and breast-cancer proliferation (Jan 2023 Q4(c), 4 marks):** oestrogen **enters the cell / nucleus** (by diffusion) · **binds a receptor / acts as a transcription factor** — *"ignore receptors cell surface membrane"* · the complex **binds the promoter / lets RNA polymerase bind** · **transcription of a gene that stimulates mitosis** · **production of a protein involved in cell division** — *"ignore proliferation"*.
- **Steroid hormones regulating transcription (Oct 2024 Q2(b)(ii), 3 marks):** **diffuses across the membrane** · **binds receptor inside cell / nucleus** — *"reject binding to cell membrane receptors"* · **acts as / interacts with a transcription factor** · **activation or repression of transcription** — *"ignore binds to DNA"*.
- **Reduced testosterone and muscle protein (Jan 2024 Q8(c), 4 marks):** **less testosterone in blood** · **less enters muscle cells / nucleus** · **fewer hormone–receptor complexes / transcription factors** · **less transcription / mRNA of muscle protein genes** — *"to get full marks they have to mention reduced / less / fewer at least ONCE or it is max 3 marks."*
- **TBT data (Oct 2024 Q2(b)(i)):** TBT **lowers testosterone** (negative correlation) · oestradiol **rises up to 50 mg kg⁻¹ then falls** · testosterone is **always higher** than oestradiol.

20.2 Peptide hormones — mark scheme chains

- **AVT (bird ADH) stimulating aquaporins (June 2026 Q4(b)(ii), 4 marks):** AVT **released into the blood** · **cannot cross the membrane** · **binds a receptor** on the collecting-duct cell · **activates a second messenger / cAMP** · second messenger **activates transcription** · **aquaporin genes transcribed / more aquaporins made**.

- **Growth hormone and muscle protein (June 2025 Q7(a)(ii), 3 marks):** GH binds a receptor on the sarcolemma · activates a second messenger / transcription factor — "*ACCEPT activates enzymes / cAMP*" · RNA polymerase binds the promoter · transcription of muscle protein genes (actin, myosin) → translation.
- **Insulin and glucose uptake (Jan 2024 Q8(e), 2 marks):** insulin binds receptors on the muscle cell membrane · increases the number of GLUT4 / glucose transport proteins in the membrane · glucose enters by facilitated diffusion.
- **Growth hormone with age (June 2025 Q7(b)):** GH decreases with age · rapid decline up to about 30 years, slower after · a specific error-bar comment · a methodology comment (sample size unknown, lifestyle, disease).

20.3 Adrenaline and "fight or flight"

- Hormone: **adrenaline** (*June 2023 Q1(b)(i); Oct 2024 Q2(a)(ii); June 2025 Q7(c)*), from the adrenal glands.
- **Role (Jan 2023 Q7(b), 3 marks):** released into the blood from the adrenal glands / binds receptors on target organs · two named effects: **increased heart rate**, **vasodilation to skeletal muscle**, **increased breathing rate**, **pupil dilation**, **glycogen → glucose in the liver**, **reduced gut activity**.
- **Statement MCQ (June 2022 Q6(a)):** "arterioles constrict in skeletal muscles" ✗ (they dilate) · "glycogen is released from the liver" ✗ (glucose is) · "rate of digestion decreases" ✓ → **1**.
- **Ageing and adrenaline (June 2022 Q6(b)(iii), 6-mark *):** three graphs → Level 3 needs **all three commented on**, the **effect in older people**, and a **discussion of significance** (error bars). Older people: slightly **greater heart-rate** rise but **smaller rise in cardiac index** above 40 $\text{ng kg}^{-1} \text{ min}^{-1}$ — explained by **weaker cardiac muscle / lower stroke volume**.

20.4 Blood glucose and negative feedback (Jan 2023 Q7(c), 6-mark *)

Credited explanation bank:

- **receptors detect a change in blood glucose**
- **β cells secrete insulin; α cells secrete glucagon** (pancreas) — into the **blood to target cells** (liver, muscle)
- **insulin binds receptors → more glucose taken up (GLUT4), glucose → glycogen (glycogenesis), more respiration**
- **glucagon → glycogen → glucose (glycogenolysis), gluconeogenesis;** glucose released to the blood
- **blood glucose returns to the set point**
- **as glucose returns to normal, insulin / glucagon secretion falls**

The scheme's rule: "*To get 6 there has to be one explanation using negative feedback*" — i.e. the **switching-off** statement in the last bullet. Pancreas as an organ (Oct 2025 Q8(b)): **different tissues**

/ **cell types** (α and β cells) release **different hormones** that **work together** to maintain blood glucose.

21. The kidney: structure

Region	Contains
Cortex	Bowman's (renal) capsules / glomeruli, proximal and distal convoluted tubules
Medulla	loops of Henle and collecting ducts
Pelvis	collects urine → ureter

Route of filtrate / a water molecule (Jan 2022 Q4(a)(iii); June 2022 Q2(a)(iii)): glomerulus → Bowman's capsule → proximal tubule → loop of Henle → distal tubule → collecting duct.

- Structures W and X (capsule, PCT) are in the **cortex** (Oct 2020 Q2(a)(i)); ultrafiltration in **W** (capsule); the loop of Henle is **Y only**.
 - Oct 2025 Q3(a): cortex **Q**, medulla **R**, pelvis **O**; ultrafiltration structures in the **cortex**.
 - June 2024 Q7(a): the vessel taking blood to the renal vein is **Q**; the structures lying in the medulla are **P, Q and U**.
 - June 2026 Q3(a): selective reabsorption at **T** (PCT); blood **into** the nephron **S** (afferent arteriole), **away R** (efferent).
 - Draw the Bowman's capsule (June 2022 Q2(a)(i), 3 marks): cup shape around the glomerulus · two labels from efferent arteriole, glomerulus, PCT, podocytes, basement membrane, squamous epithelium.**
 - Magnification of a nephron diagram (Jan 2026 Q4(a)(i)): 5.6 (2 s.f.).**
-

22. Ultrafiltration

22.1 The chain

- The **afferent arteriole is wider than the efferent arteriole** → **high hydrostatic pressure in the glomerulus** (Jan 2025 Q3(a)(i)).
- Small molecules are **forced out** through the **pores (fenestrations) in the capillary endothelium**, the **basement membrane** and between **podocytes**.
- The **basement membrane** is the filter that **stops plasma proteins** (June 2023 Q5(a)(ii)).
- Filtered: **water, glucose, amino acids, urea, salts / ions, creatinine**. Not filtered: **blood cells, plasma proteins (haemoglobin, prothrombin)** — "*too big a molecule to be filtered*".

MCQs: filtered = **glucose and sodium chloride** (June 2022), **glucose and urea** (June 2023).

Urea (Oct 2020 Q2(b), 2 marks): forced out by **high pressure** · caused by the **afferent vessel** having a greater diameter than the efferent — *"ALLOW arteriole NOT artery"* · through **pores in the basement membrane / podocytes / fenestrations**.

Creatinine (Oct 2021 Q4(c)(i), 3 marks): small molecule · forced through the pores / basement membrane · by **high pressure in the glomerulus**. $GFR: 1960 \div 15 = 131 \text{ mg h}^{-1}$ (3 s.f.).

23. Selective reabsorption in the proximal tubule

23.1 The glucose chain (June 2022 Q2(b); June 2026 Q3(a)(iii); Oct 2024 Q6(d)(i))

1. Occurs in the **proximal convoluted tubule**.
2. **Sodium ions are actively transported out of the tubule cells into the blood** (basal membrane, Na^+/K^+ pump, ATP).
3. This keeps Na^+ **low inside** the cell, so Na^+ **diffuses in from the lumen** through a **co-transport protein**, carrying **glucose with it** — **sodium co-transport** (MCQs June 2023, Oct 2024).
4. Glucose then moves into the **blood by (facilitated) diffusion** down its concentration gradient.

"ignore ref to sodium pump" (June 2022 — where it was not linked to the gradient) · *"ACCEPT channel proteins / carrier proteins"*.

23.2 Adaptations of PCT cells (June 2021 Q4(b)(ii); June 2024 Q7(a)(iii))

microvilli (brush border) on the lumen side → **large surface area** · **many mitochondria near the base** → **ATP for active transport** — *"ignore producing energy"*

The June 2024 scheme also credits **orientation**: microvilli face the **lumen**, mitochondria are at the **basal side**.

23.3 Reading a nephron data table (June 2021 Q4(d), 3 marks)

Glucose: 1.50 in plasma and filtrate → **0.00** in urine = **100 % reabsorbed** in the **proximal tubule** by **carrier proteins**. Urea: 0.30 → **21.00** = **70× more concentrated** because **water is reabsorbed** (and some urea is secreted).

23.4 The renal threshold for glucose (Jan 2026 Q4(b))

- Filtration rate **rises in proportion** to plasma glucose ($0.5 \rightarrow 3.5 \text{ mmol min}^{-1} = 600 \text{ % increase}$).
- Reabsorption rises until about **23–25 mmol dm⁻³**, slowing from **13–15**, then **plateaus**.
- **Glucose appears in urine above 12 mmol dm⁻³** because **filtration is faster than reabsorption / the carriers are saturated** (reabsorption at maximum).

23.5 Where amino acids come from, and urea

- Most amino acids reabsorbed in the **proximal tubule** (*Jan 2022 Q4(a)(ii)*).
 - **Urea is made in the liver from excess amino acids** (*June 2022 Q2(c)*).
 - **Fate of excess amino acids (Jan 2025 Q1(b), 3 marks)**: transported to the **liver** · **deamination** — **amine group removed** — *"IGNORE amino group"* · amine group **converted to urea** · in the **ornithine cycle**; the remainder → keto acid / glucose / pyruvate.
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24. The loop of Henle: the countercurrent multiplier

24.1 The chain (*Jan 2022 Q4(b)*, 4 marks; *June 2023 Q5(b)*, 4 marks)

1. Filtrate in the two limbs flows in **opposite directions** (countercurrent).
2. The **ascending limb actively transports Na^+ and Cl^- out** into the medulla — *"do not accept sodium"* (write ions); *"can get [the active transport mark] even if the ions were pumped in the wrong direction."*
3. The ascending limb is **impermeable to water**.
4. The medullary tissue fluid therefore has a **high solute concentration / low water potential** — *"ACCEPT (surrounding) fluid becomes hypertonic"*.
5. **Water leaves the descending limb by osmosis** (through aquaporins).
6. The effect is **multiplied down the loop**, so the **deepest medulla has the lowest water potential**.
7. Water is drawn **out of the collecting duct by osmosis** as it passes through the medulla → **concentrated urine**.

24.2 Desert mammals — kangaroo rat, camel, fennec fox

Credited points (*Oct 2020 Q2(c)*; *Oct 2022 Q5(d)*; *Jan 2024 Q7(b)*; *Oct 2025 Q3(b)(ii)*; *June 2026 Q3(b)(ii)*):

longer loops of Henle (more juxtamedullary nephrons, **thicker medulla**) · **more Na^+ / Cl^- pumps** in the ascending limb → **lower water potential in the medulla** · **greater countercurrent effect** · **more aquaporins / more ADH sensitivity** in the collecting duct · **more water reabsorbed by osmosis** → **very concentrated urine** · **more mitochondria in the thick ascending limb** to make **ATP for active transport** · behavioural: **nocturnal / burrows by day**

- **Kangaroo rat vs brown rat (Oct 2020 Q2(c), 3 marks)**: less water in deserts · **more concentrated urine** · must **actively transport more Na^+ into the medulla** · so needs **more mitochondria for ATP** — *"IGNORE sodium"*.
- **Camel MCQ (Oct 2022 Q5(a))**: concentrated urine ✓, long loop ✓; high GFR ✗, reduced ADH ✗ → 2.

- **Medulla : cortex ratio (June 2026 Q3(b)(i)):** meerkat (desert) — **any value between 4.7 : 1 and 7.0 : 1.**
 - **Habitat vs urine concentration (Oct 2025 Q3(b)(i)):** **the wetter the habitat, the more dilute the urine** · **smaller animals produce more concentrated urine** than larger ones in the same habitat.
 - **Water balance kangaroo rat vs human (Jan 2024 Q7(c)):** difference in urine over 7 days: $10\,500 - 3.15 = 1.05 \times 10^4 \text{ cm}^3$ — "*no units max 1 mark*". Comment: the rat gains **most water from metabolism** (it drinks nothing) and loses **very little in urine** (long loop, concentrated urine) — each comparison must be **linked to an explanation**.
 - **Water use (Oct 2022 Q5(b)):** buffalo 150×697 – camel $57 \times 850 = 56\,100 \text{ cm}^3 = 56 \text{ dm}^3$ (2 s.f.). No correlation with mass because of **different adaptations / habitats**.
 - **Kangaroo rats on fresh vs sea water (Oct 2022 Q5(c)):** both **increase mass**, fresh water **more** · a **calculated change** (e.g. 6 g difference between fresh and no water).
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25. ADH and osmoregulation

25.1 The chain

1. **Osmoreceptors in the hypothalamus** detect a **fall in the water potential of the blood** (*June 2022 Q3(a)(iv)*: "osmoreceptors"; *Oct 2024 Q6(b)(i)* MCQ).
2. ADH is **made in the hypothalamus** (*June 2022 Q3(a)(i)*: answer **B hypothalamus** — "D is not correct as the pituitary gland does not produce ADH") and **released from the posterior pituitary** into the **blood**.
3. ADH acts on the **collecting duct** (and DCT) — *June 2021 Q4(a)*: of Bowman's capsule, collecting duct, loop of Henle and proximal tubule, **only one** responds; *June 2022 Q3(a)(ii)*: **collecting ducts**.
4. ADH **binds receptors** on the collecting-duct cells → vesicles containing **aquaporins fuse with the (apical) membrane**.
5. The walls become **more permeable to water**.
6. **More water is reabsorbed by osmosis** into the blood → **small volume of concentrated urine**.
7. Blood water potential returns to normal → **less ADH released** (negative feedback). The whole process is **negative feedback** (*June 2022 Q3(a)(iii)*).

Asked in *Jan 2022 Q4(c)* (4), *Jan 2025 Q3(c)* (3), *June 2024 Q7(c)* (3), *Oct 2024 Q6(c)* (4), *June 2023 Q5(c)(iii)* (2), *Jan 2025 Q8(d)* (5).

Rulings:

- "*ignore permeability of tubules decreases unqualified (must say a correct named tubule)*" (*Oct 2024*). Name the **DCT / collecting duct**.
- "*ignore more urine produced*" — say **more water excreted / more dilute urine** (*Oct 2024*).

- "ignore loop of Henle" as the ADH target (June 2023).
- "Accept insertion of aquaporins into cell membranes (of CD and DCT)" (Jan 2022).

25.2 The reverse case — drinking too much water (Oct 2024 Q6(c), 4 marks)

osmoreceptors detect the fall in solute concentration · baroreceptors in the carotid artery / aortic arch detect raised blood volume/pressure · the pituitary releases less ADH · fewer aquaporins in the DCT / collecting duct · less water reabsorbed → dilute urine

Why plasma concentration must be held in a narrow range (Oct 2024 Q6(b)(ii)): to prevent water entering or leaving cells by osmosis — cells would lyse or crenate.

25.3 ADH going wrong

- **Hyponatraemia from excess ADH (June 2024 Q7(c), 3 marks):** ADH binds receptors on DCT / collecting ducts · aquaporins move into the apical membrane · more water reabsorbed · the blood becomes more dilute — its sodium ion concentration falls.
- **MDMA and brain swelling (Oct 2020 Q3(b), 6-mark *):** as MDMA rises, ADH rises · 9 h after MDMA, ADH still ~3× the 96 h value · as ADH rises plasma Na⁺ falls · low Na⁺ goes with brain swelling · ADH → more water reabsorbed → blood diluted → water moves by osmosis into brain tissue → swelling. This is a "comment on" question marked with the synthesis / judgement grid: "Level 3 ... Most of the important or relevant scientific factors ... are synthesised. A detailed and accurate judgement is made."
- **Keeping sodium ions in range (Jan 2025 Q8(d), 5 marks):** by homeostasis / osmoregulation · osmoreceptors in the hypothalamus detect water potential · impulses to the pituitary · more or less ADH ("ACCEPT aldosterone") · DCT / collecting duct reabsorb more or less water / sodium ions.
- **Cells shrink when blood electrolytes rise (Jan 2025 Q8(e)):** water leaves the cell by osmosis · because the water potential is lower outside (higher solute concentration in the blood / tissue fluid).

25.4 Reading urine data

- **ADH vs urine (Jan 2025 Q3(b)):** as ADH rises, urine concentration rises and urine production falls. At 75 % ADH: $1.5 \times 60 \times 24 = 2.16 \times 10^3 \text{ cm}^3 \text{ day}^{-1}$. Blood filtered in 24 h at $6000 \text{ cm}^3 \text{ h}^{-1} = 144 \text{ dm}^3$ (MCQ).
- **After drinking 1000 cm³ (June 2024 Q7(b), 4 marks):** urine volume peaks at 60 min · ion concentration lowest at 60 min · urea mass peaks at 60 min · by 150 min all return to the start values · linkage (volume highest when ion concentration lowest) · no sample size / error bars comment.
- **Nothing / water / salt solution (June 2023 Q5(c)):** ratio for females 1 : 8 : 4 (from 100 : 840 : 440, 1 s.f.) · drinking increases urine output · the increase is greater in females · drinking

water raises blood water potential → less ADH → collecting duct less permeable → less water reabsorbed.

- **Daily urine (June 2021 Q4(e)):** $0.83 \times 60 \times 24 = 1195 \text{ cm}^3 = \mathbf{1.2 \text{ dm}^3}$ (2 s.f.).
 - **Renal vein area (June 2021 Q4(c)):** $r = 2.355 \text{ mm} \rightarrow \pi r^2 = \mathbf{17.4 \text{ mm}^2}$; difference from 0.194 cm^2 ($= 19.4 \text{ mm}^2$) = $\mathbf{1.97 \text{ mm}^2}$.
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PART D — NERVES, SYNAPSES, BRAIN, EYE AND BEHAVIOUR

26. Neurones and the reflex arc

26.1 Three neurones

	Sensory	Relay	Motor
Carries impulses	from receptor to CNS	within the CNS (sensory → motor)	from CNS to effector (muscle / gland)
Cell body	in the dorsal root ganglion , part-way along (<i>Jan 2022 Q3(a)(ii)</i> : "in the middle of the axon")	in the grey matter	in the grey matter , at the end of the axon

Compare sensory and motor (June 2022 Q5(a)(ii), 3 marks): similarities — **both transmit impulses** · both **release neurotransmitter / synapse with relay neurones**; differences — sensory carries impulses **towards the CNS**, motor **away from the CNS** · sensory **from receptors**, motor **to effectors**. *"do not piece together – has to be in same / adjacent sentences"; "do not accept messages / signals"; "ignore brain"*. Oct 2024 Q1(c) (2) wants two **functional** differences, mix-and-match allowed.

26.2 Labelling

- Relay neurone (*Oct 2022 Q3(a)(i)*): L **dendrites**, M **cell body**, N **axon**, O **presynaptic knob / axon terminal** — *"do not accept synapse(s)"*.
- Motor neurone (*June 2025 Q1(a)*): A **dendrite(s)**, B **cell body** — *"IGNORE body cell"*, C **myelin sheath** — *"ACCEPT Schwann cell"; "IGNORE lipid unqualified / sheath unqualified"*.
- Draw a myelinated motor neurone (June 2024 Q5(c)(i), 3 marks):** **myelin sheath with nodes of Ranvier** · **cell body at one end, synaptic terminals at the other** · two correct labels — *"label arrows MUST BE touching"; "max 2 if sensory / relay neuron drawn"*.
- Direction of impulse (*Oct 2024 Q1(a)*): one arrow, one direction — *"reject arrows in 2 different directions"*.

26.3 The spinal reflex arc

receptor → sensory neurone (cell body in dorsal root ganglion) → relay neurone in the grey matter of the spinal cord → motor neurone → effector

- **Describe the structure (Jan 2023 Q3(b), 3 marks):** receptor cells synapse with the **sensory neurone** · which synapses with a **relay neurone in the grey matter** · relay synapses with **motor neurone** · motor neurone synapses with the **effector**.
 - **Skin to CNS (Jan 2024 Q3(a)(ii)):** receptors synapse with **sensory neurones** · impulse **towards the spinal cord** · through the **dorsal root ganglion** · synapse with **relay / motor neurone** · in the **grey matter**.
 - **MCQs:** component immediately after the receptor = **sensory neurone** (*June 2021*); reflex pathway = **receptor** → **spinal cord** → **muscle**; *Jan 2026 Q3(a)*: J **motor**, K **sensory**, L **relay**; grey matter = the central region Q.
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27. The resting potential

27.1 The chain (Oct 2020 Q6(b), 4 marks; June 2021 Q2(b)(i), 3 marks)

1. The **sodium–potassium pump** actively transports **3 Na⁺ out and 2 K⁺ in**, using **ATP**.
2. This sets up **concentration gradients: high K⁺ inside, high Na⁺ outside**.
3. At rest the **voltage-gated Na⁺ channels are closed**, and the membrane is **more permeable to K⁺** (K⁺ leak channels).
4. **K⁺ diffuses out**, so the **inside is negative** relative to the outside — about **–70 mV**.
5. The pump keeps the potential difference **stable**.

Why maintaining the resting potential needs ATP (Jan 2021 Q1(b), 3 marks): ATP used by the **sodium–potassium pump** · to move **Na⁺ and K⁺ across the membrane** · to **establish concentration gradients** for both ions.

Ion ratio MCQ (Jan 2025 Q4(c)): K⁺ outside : inside = 4 : 155 = **0.026 : 1**.

28. The action potential

28.1 The sequence

1. A stimulus opens **some sodium ion channels; Na⁺ diffuses in**, making the inside less negative.
2. If the **threshold** (about –55 mV) is reached, **voltage-gated Na⁺ channels open — positive feedback**, more Na⁺ floods in.
3. **Depolarisation** to about **+40 mV** — the inside is now positive.
4. **Na⁺ channels close; voltage-gated K⁺ channels open; K⁺ diffuses out** → **repolarisation**.
5. K⁺ channels are slow to close → **hyperpolarisation** (below –70 mV).
6. The **refractory period**: no new action potential can be generated. This makes impulses **discrete and one-way**.

7. The Na⁺/K⁺ pump restores the resting potential.

Action potentials are **all-or-nothing**: a stronger stimulus gives **more frequent** impulses, not bigger ones.

28.2 The MCQs — every paper has one

Paper	Question	Answer
Jan 2021 Q1(a)	resting potential; channels open at the falling phase Y; state at Z	-70 mV · voltage-gated K⁺ only · hyperpolarised
Jan 2021 Q1(a)(iv)	"magnitude proportional to stimulus", "spread both directions", "refractory period"	one correct — only the refractory period
Jan 2023 Q3(a)	ion movement that produces an action potential; repolarisation	Na⁺ into the axon ; Na ⁺ channels closed , potential decreasing
Jan 2024 Q2(a)(i)	depolarisation	Na ⁺ channels open , potential increasing
Oct 2024 Q1(b)	resting potential; hyperpolarised point; V to W difference	-70 mV · W · 140 mV
Oct 2025 Q6(a)(i)	cell producing myelin	Schwann cell (accept oligodendrocyte)

28.3 Short explain questions

- **Repolarisation (Jan 2024 Q2(a)(ii), 3 marks):** Na⁺ (voltage-gated) channels close · K⁺ move out through voltage-gated K⁺ channels · the inside becomes negative relative to the outside — *"reject contradictions e.g. potassium ion moving in"; "ignore back to resting potential"*.
- **Producing an action potential (Jan 2025 Q4(a), 4 marks):** neurotransmitter binds · Na⁺ channels open · Na⁺ diffuse in · positive feedback, more channels open · depolarisation · threshold reference.
- **Membrane potential rising from -70 to -55 then 0 (Jan 2022 Q6(b)):** stimulus changes the membrane potential · Na⁺ channels open, Na⁺ rush in, making it less negative · above threshold → depolarisation initiated.
- **Temperature and conduction speed (Jan 2025 Q4(d)(i)):** speed increases · ions diffuse faster / channels open and close faster / ions have more kinetic energy — *"IGNORE enzyme activity"*.
- **Maximum rate of increase of K⁺ permeability (Jan 2025 Q4(b)):** steepest gradient $9 \div 0.5 = 18$ a.u. ms⁻¹ (2 s.f., with units).

28.4 Generator potentials: the Pacinian corpuscle (June 2021 Q2(b))

- **How pressure causes an action potential (3):** pressure **deforms the membranes / lamellae** · **stretch-mediated Na⁺ channels open, Na⁺ floods in** → generator potential · detail of the action potential in the axon.
- **Why electrode Y reads +40 mV for both medium and high pressure (3):** the **threshold has been reached** · an **action potential** results · action potentials are **all-or-nothing**.

29. Myelination and saltatory conduction

29.1 The three-point chain — asked in five papers

June 2021 Q3(a), Jan 2022 Q6(c)(ii), Oct 2022 Q3(b)(iii), Jan 2024 Q2(b)(iii), Oct 2025 Q6(a):

myelin is an (electrical) insulator — ions cannot cross the membrane where it is myelinated · **action potentials occur only at the nodes of Ranvier**, where voltage-gated Na⁺ channels are concentrated · the impulse **jumps from node to node** — **saltatory conduction** — so it travels faster than along a non-myelinated axon of the same diameter

- *"ignore myelinated neurone {has / creates} insulation"* (Jan 2024) — say **myelin is an insulator**.
- Oct 2025: saltatory conduction = impulse **jumps from one node to another** / depolarisation **only at nodes**; the property = **insulator / impermeable to ions**.
- MCQ Oct 2020 Q6(a)(ii): myelinated is **faster** because depolarisation occurs **at the nodes of Ranvier**.

29.2 Axon diameter data

- Jan 2025 Q4(d)(ii): **larger diameter** → **faster** · **myelination increases speed** · **myelination has a greater effect than diameter**.
- June 2026 Q6(a): **myelinated faster** · for non-myelinated, **greater diameter** → **faster** · for myelinated, **no correlation** with diameter (human 15 μm → 120 m s⁻¹; cat 10 μm → 50).
- Jan 2022 Q6(c)(i): gradient of the myelinated line 120 ÷ 20 = 6; % difference from 0.5 = **1100 %**.

29.3 Demyelinating diseases — one chain, four contexts

Paper	Disease	Credited points
June 2021 Q3(b)	Polyneuropathy → muscle weakness	transmission impaired / slower · fewer impulses reach the muscle · fewer muscle fibres stimulated

Paper	Disease	Credited points
June 2022 Q5(b)	Guillain-Barré	loss of insulating myelin · slower impulses — no saltatory conduction · motor neurones transmit fewer / slower impulses → loss of function · sensory impulses do not reach the CNS → no sensation — <i>"ignore impulses being lost"</i>
June 2023 Q8(f)	Multiple sclerosis	loss of myelin / nodes of Ranvier · slower conduction / reduced saltatory conduction
Oct 2023 Q3(c)	ADEM → problems walking	loss of myelin in motor neurones · lesions in the cerebellum · fewer impulses reach leg muscles

Myelin and dementia validity (June 2021 Q3(c)): all dementia types have **less myelin** than the control · **no relationship between myelin and the type** of dementia · **SDs do not overlap** with the control, which **supports validity**.

30. Synaptic transmission — the most-asked chain in the nervous system

30.1 The chain

1. The action potential arrives at the **presynaptic knob**.
2. **Voltage-gated calcium ion channels open**; Ca^{2+} **diffuse in** (Jan 2026 Q3(b)(i) MCQ: calcium ion channels).
3. Ca^{2+} cause **vesicles containing neurotransmitter to move to and fuse with the presynaptic membrane**.
4. Neurotransmitter is **released by exocytosis** into the **synaptic cleft**.
5. It **diffuses across** the cleft and **binds to (complementary) receptors on the postsynaptic membrane**.
6. **Ligand-gated sodium ion channels open**; Na^+ **enter**; the postsynaptic membrane **depolarises**.
7. If the **threshold** is reached, an **action potential** is generated in the postsynaptic neurone.
8. The neurotransmitter is **broken down** (acetylcholine by **acetylcholinesterase**) or **taken back up**, so the signal stops.

Asked in Jan 2021 Q7(a)(i) (4), Jan 2023 Q3(c) (4), Oct 2023 Q5(a) (4), June 2025 Q1(b) (3), Oct 2025 Q6(b) (3), June 2026 Q6(b)(i) (3). Neurotransmitters named in MCQs: **acetylcholine**, **dopamine**, **adrenaline** and **acetylcholine** (Jan 2022, Jan 2025, Jan 2026).

Rulings: "IGNORE sodium ion channel as a receptor" · "ignore cation channels" · "ignore if the calcium ions are going into the wrong structure" (Jan 2023 — i.e. no credit for them) · "ACCEPT binds as eq fuse" · "ACCEPT exocytosis of neurotransmitter" · "IGNORE action potential" where depolarisation is asked (June 2026).

The space between neurones is the **synaptic cleft** (June 2022 Q5(a)(i)).

30.2 Why different neurones respond to different transmitters (June 2022 Q8(b), 3 marks)

receptors on the postsynaptic membrane · have **different 3D shapes / are complementary to specific neurotransmitters** · describe what happens on binding — **Na⁺ channels open / depolarisation**; an **inhibitory** transmitter **prevents depolarisation**

30.3 Inhibitory synapses (Oct 2019 Q7(g), 5-mark QWC)

Diazepam: **binds receptors on the postsynaptic membrane** · **opens chloride ion channels / Cl⁻ move in** · the inside becomes **more negative (hyperpolarised)** · postsynaptic membrane **not depolarised** · action potential **less likely**.

30.4 Calcium channel mutations (June 2022 Q5(c), 4 marks)

mutation changes the **base sequence** → **amino acid sequence** of the channel protein · channel **changes shape / cannot open** · **no Ca²⁺ influx** into the presynaptic knob · **fewer vesicles fuse**, less neurotransmitter released · **fewer action potentials** in the postsynaptic neurone

31. Drugs and toxins at synapses and axons

Drug / toxin	Action	Effect	Papers
Nicotine	similar shape to acetylcholine — binds ACh receptors	stimulates — more action potentials	Jan 2024 Q4(b); Oct 2023 Q2(b)(ii); Jan 2026 Q1(a)
Cobra venom α-toxin (CV)	binds ACh receptors but does not open them	blocks transmission	Jan 2026 Q1(a)
Lidocaine (local anaesthetic)	blocks / closes voltage-gated Na⁺ channels	no depolarisation, no impulses	Oct 2022 Q3(a)(ii); June 2025 Q3(a); Jan 2025 Q6(b)

Drug / toxin	Action	Effect	Papers
Tetrodotoxin (TTX)	binds and blocks voltage-gated Na⁺ channels	membrane cannot reach threshold → no action potential	Oct 2020 Q6(c); June 2024 Q5(c)(ii); June 2025 Q3(c)
MDMA (ecstasy)	increases release of serotonin (and blocks reuptake)	increases transmission	June 2025 Q3(b); June 2023 Q7(c); Oct 2023 Q2(b)(ii)
SSRIs	block the reuptake proteins for serotonin	serotonin stays in the cleft → more depolarisation	June 2024 Q2(b)(ii)
L-DOPA	crosses the blood–brain barrier , converted to dopamine	replaces lost dopamine	Oct 2020, Jan 2021, June 2022, Oct 2024, June 2025
Acetylcholinesterase inhibitors	ACh not broken down	ACh keeps stimulating receptors	June 2026 Q6(b)(iii)
Caffeine	crosses the BBB; blocks adenosine receptors	more depolarisation / neurotransmitter release	Jan 2025 Q8(b)

Two statement MCQs: Oct 2023 Q2(b)(ii): drugs that **increase** transmission = **MDMA and nicotine** (lidocaine decreases it). Jan 2026 Q1(a): "binds to ACh receptor" — **both nicotine and CV**; "stimulates nerve impulses" — **nicotine only**.

31.1 The chains

- **TTX blocks an impulse (June 2024 Q5(c)(ii), 3 marks):** voltage-gated **Na⁺ channels do not open / are blocked** · **Na⁺ cannot enter** — "*do not accept sodium unqualified*" · membrane cannot depolarise / reach threshold · **no action potential** — "*ignore fewer action potentials*".
- **TTX and breathing (June 2025 Q3(c), 3 marks):** blocks **Na⁺ (or Ca²⁺) channels** · **no depolarisation / action potentials** · **no impulses to the intercostal muscles / diaphragm** (or from the **respiratory centre**).
- **Nicotine in knockout mice (Jan 2024 Q4(b), 4 marks):** nicotine **increases action potentials** in the postsynaptic neurone · because it **binds ACh receptors** — "*ignore nicotine blocks receptors*" · group C (no receptor gene) is **like the no-nicotine group** · because there is **no ACh receptor** on the postsynaptic membrane.
- **SSRIs and depression (June 2024 Q2(b)(ii), 3 marks):** depression linked to **lack of serotonin** · SSRIs **bind the reuptake proteins** · **serotonin reuptake is inhibited** — stays in the cleft · serotonin **binds postsynaptic receptors** · **more depolarisation / action potentials**.

- **MDMA by mouth (June 2023 Q7(c)(iii))**: absorbed into the blood from the gut · crosses the blood–brain barrier · binds neurones / receptors · stimulates serotonin release.
- **Acetylcholinesterase inhibitors in myasthenia gravis (June 2026 Q6(b)(iii))**: ACh not broken down / stays in the cleft · keeps stimulating the remaining receptors · Na⁺ channels stay open · threshold reached / impulses generated.
- **Myasthenia gravis weakness (June 2026 Q6(b)(ii))**: fewer ACh receptors → less chance of reaching threshold · fewer impulses to the muscle · less forceful contraction — *"REJECT 'no'"*.
- **Lidocaine protecting heart cells in low oxygen (Jan 2025 Q6(b)(iii))**: prevents depolarisation / Na⁺ influx · reduces heart-muscle contraction · so less need for oxygen / ATP — *"ACCEPT reduced lactic acid"*. Data (6(b)(i)): low O₂ reduces survival · higher lidocaine → higher survival · a specific error-bar comment.
- **Blocking K⁺ channels (June 2019 Q7(e))**: K⁺ cannot leave · membrane stays depolarised / cannot repolarise · impulses continuously generated.
- **α-latrotoxin (June 2019 Q7(g))**: Ca²⁺ → vesicles move to / fuse with presynaptic membrane → release by exocytosis.

31.2 Dose–response curves

- **LD₅₀ = 8.4 µg kg⁻¹ (June 2025 Q3(d)(i))**: the dose that kills 50 % of the mice — *"Both parts needed"*.
- **Toxicity comment (3(d)(ii))**: no deaths below 6 µg kg⁻¹ · deaths increase above 6 · all mice die at 10 · methodology — age / sex / strain / number of mice; no error bars.
- **Log₁₀ scale (Jan 2026 Q1(b)(i))**: fits a wide range of concentrations on one axis · makes the trend / rate of change easier to see — *"Ignore easier comparison"*.
- **ED₅₀ difference (Jan 2026 Q1(b)(ii))**: read 3.5 and 3.8 on the log axis → antilogs 3162 and 6310 → difference 3147. *"1 mark for 3.8 – 3.5 = 0.3"* — subtracting logs is the trap.

32. Parkinson's disease, dopamine and depression

32.1 The chain

- Parkinson's = death of dopamine-producing neurones in the substantia nigra / midbrain.
- **Less dopamine released** → fewer dopamine molecules bind receptors / ligand-gated Na⁺ channels on the postsynaptic membrane → fewer action potentials → fewer impulses to the brain regions controlling movement / muscles → tremor, rigidity, loss of motor control (Oct 2020 Q7(a)(ii); Oct 2024 Q4(a)).
- **Why symptoms increase with age (Oct 2024 Q4(a), 4 marks)**: dopamine cells decrease with age · after 44–45 years they fall below 350 000 · less dopamine released — *"ignore {no / lack of} dopamine"* · fewer impulses / action potentials · reduced muscle control.

32.2 L-DOPA — the chain, and its two add-ons

L-DOPA (June 2025 Q4(a), 4 marks): absorbed into the blood from the gut · crosses the blood–brain barrier · converted to dopamine in the brain · replacing the dopamine lost by the death of dopamine-producing neurones — *"Need BOTH parts"*.

- MCQ Oct 2020 Q7(a)(i): L-DOPA crosses the BBB and is then converted to dopamine in the brain.
- **Dopamine cannot cross the BBB** (June 2024 Q2(b)(i)): "dopamine cannot pass from bloodstream into neurones in the brain").
- **Co-beneldopa / carbidopa (June 2022 Q4(a); Oct 2024 Q4(b)):** dopamine cannot cross the BBB · the inhibitor stops L-DOPA being converted to dopamine in the gut / blood · so more L-DOPA reaches the brain · lower doses can be used. *"L-Dopa shouldn't be converted to dopamine until after it has crossed the blood brain barrier = 2 marks."*

32.3 Research questions

- **Release of dopamine could be inhibited by (Oct 2020 Q7(b)):** less Ca^{2+} influx → vesicles do not fuse; or changed membrane proteins preventing fusion.
 - **Dopamine and Drosophila heart rate (Oct 2024 Q4(c)):** heart rate rises then falls as concentration increases · peak at log 3–4 · **specific error-bar comment** (e.g. log 0–2 overlap → no significant difference) · **only 10 flies / one species**.
 - **Dendrites in LRRK2 neurones (Jan 2021 Q7(b)(ii)):** number and length of dendrites lower in Parkinson's · error bars do not overlap with controls → significant · **overlap between the two Parkinson's groups** → no significant difference.
 - **Transcription factors and dopamine synthesis (Jan 2021 Q7(b)(i)):** TFs bind the promoter of genes for dopamine synthesis · allow mRNA synthesis / RNA polymerase to bind · mRNA translated into the enzymes that synthesise dopamine.
 - **Food toxicants and Parkinson's (Jan 2022 Q8(a)):** absorbed into the blood · cross the BBB · kill dopamine-producing cells / reduce dopamine secretion.
 - **Serotonin statements (June 2023 Q7(a)):** "L-dopa converts to serotonin" ✗; "serotonin is a neurotransmitter" ✓; "made by postsynaptic neurones" ✗ → 1.
 - **Dopamine and sucrose in rats (Jan 2023 Q7(a)(i)):** positive correlation · leads to greater desire to eat · overlapping error bars → not significant · correlation, not causation · methodology (number of rats, how "desire" was measured). SD (Q7(a)(ii)): $\Sigma(x - \bar{x})^2 = 434$, $n - 1 = 3$ → **S = 12** (2 s.f.).
-

33. Brain regions

Region	Function the schemes accept	Rejections
Cerebral hemispheres / cerebrum / cortex	voluntary actions / movement, learning and memory, thinking, intelligence, language, interpreting information from the retina (visual cortex), emotions	—
Cerebellum	coordination of voluntary movement, balance, posture	"ignore co-ordination unqualified"
Medulla oblongata	control of heart rate (cardiovascular centre), breathing (ventilation centre), blood pressure, non-conscious / autonomic processes	"release of ADH" X (June 2025 Q4(b))
Hypothalamus	thermoregulation, osmoregulation (osmoreceptors), makes ADH, controls the pituitary, homeostasis	—
Pituitary gland	releases ADH (posterior), growth hormone, regulates growth / metabolism	—

Labelling / function questions: Oct 2019 Q2(a) (P cerebral cortex, Q cerebellum, R medulla, S hypothalamus), Jan 2021 Q3(a), Oct 2021 Q8(a) (ADH secretion: **posterior pituitary** — "Accept hypothalamus"), June 2022 Q1, Jan 2023 Q2(c), June 2024 Q2(a) (cardiovascular centre **P = medulla**; balance **T = cerebellum**; solute concentration **Q and R = pituitary and hypothalamus**), Jan 2025 Q2, June 2025 Q7(a)(i) (pituitary **J**), Oct 2025 Q5(b)(iv) (learning and memory = **cerebral hemispheres**).

34. Brain imaging: CT, MRI, fMRI and PET

	How it works	Shows	Pros	Cons
CT	X-rays + computer → cross-sectional image of a thin slice (Jan 2021 Q3(b)(i))	structure only (tumours, bleeds, plaques)	quick, cheap	ionising radiation ; lower soft-tissue detail; not activity
MRI	magnetic field and radio waves → 2D / 3D image	detailed soft tissue structure	no ionising radiation ; high soft-tissue contrast	expensive, slow, noisy, enclosed ; not with pacemakers / metal implants ; does not show activity

	How it works	Shows	Pros	Cons
fMRI	active areas have more blood flow / oxyhaemoglobin ; oxyhaemoglobin and deoxyhaemoglobin absorb radio waves differently	brain activity in real time — <i>"study changes in brain function as they happen"</i> (Jan 2023 Q2(a))	no radiation; function and structure	expensive; person must stay still
PET	radioactive tracer (e.g. glucose with a positron emitter , carbon-11 / oxygen-15) → active cells take up more → positrons produce gamma rays detected → image	metabolic activity / biochemical change	shows which areas are active ; diagnoses dementia, cancer	radiation exposure ; lower resolution; expensive

The mark-scheme chains:

- **fMRI for decision-making in squash (Oct 2019 Q2(b), 3 marks):** simulate the task (video / imagine hitting the ball) · regions involved are **stimulated** · they receive **more blood / oxygen / oxyhaemoglobin** · **oxyhaemoglobin does not absorb / deoxyhaemoglobin absorbs radio waves**.
- **PET for a seizure (Jan 2021 Q3(b)(ii), 3 marks):** isotope that emits positrons / is incorporated into glucose · more active neurones respire more · so take up more glucose · positrons → gamma rays detected → image.
- **PET before and after cocaine (Jan 2022 Q6(d), 4 marks):** tracer with glucose injected · taken up by active respiring cells · scanner detects radioactivity / gamma rays · builds 2D/3D images · more regions take up radioactivity after cocaine.
- **fMRI vs CT for plaques (Oct 2021 Q8(b), 4 marks — 2 similarities + 2 differences):** both **high-resolution neuro-imaging** · both **image plaques**; fMRI **expensive**, CT cheaper · fMRI shows **activity / oxygen uptake**, CT **only structure** · fMRI uses **magnetic resonance**, CT **X-rays** — *"ignore fMRI has higher resolution than CT"*.
- **PET vs MRI pros and cons (Jan 2023 Q2(b)):** one advantage and one disadvantage of each (see table) — *"ignore safe for all individuals"*.
- **PET vs CT (June 2024 Q5(b)(ii)):** both **produce images**, both **non-invasive**; CT **X-rays** vs PET tracers / gamma rays; CT **quicker / cheaper**; PET shows **metabolic activity** — *"ignore references to 2D / 3D"*.
- **MRI for a torn ACL (June 2024 Q1(a)(ii)):** radio waves / magnetic field · 2D / 3D image — *"ignore image unqualified", "ignore high resolution image"*.
- **Combined MRI + PET (Oct 2024 Q8(c)):** 3D image · detailed structure from MRI + **areas of activity / glucose uptake / amyloid** from PET.
- **CT to measure muscle size (Jan 2024 Q8(f)):** X-rays → **cross-sectional image** · compare size **before and after** the exercise programme.

- **Dye in an MRI scan (Oct 2023 Q3(b)):** to **enhance the image / increase contrast** so the tumour is easier to locate.
- **fMRI with probiotics (June 2023 Q8(d)):** scan **while / after** eating probiotics · **compare activity** in memory and sensation regions (active areas lighter).
- **Detecting brain activity (June 2026 Q8(c)):** fMRI — increased **oxygen / oxyhaemoglobin / blood flow** in active regions; or PET — **radioactive glucose** taken up by active regions ("ACCEPT EEG"; "IGNORE CT scan").

Lesion calculations (Oct 2023 Q3(a)): $V = \frac{4}{3} \pi (0.985)^3 = 4.00 \text{ cm}^3$ (2 d.p.; accept 3.94–4.06);
actual width = $(5.5 \div 12) \times 1.97 = 0.90 \text{ cm}$ (2 s.f.).

35. The eye: rods, rhodopsin and the pupil reflex

35.1 Rod cell structure

Outer segment (stacked membranes containing **rhodopsin**, facing **away from the light** — Oct 2019 Q6(a)(i)) · **inner segment** with many **mitochondria** (the "structure that provides energy", June 2026 Q2(b)(i): K) · **nucleus** · **synaptic region** facing the light, synapsing with a **bipolar neurone**.

35.2 Rods in the dark and in the light

	In the dark	In the light
Rhodopsin	intact: opsin + cis-retinal	absorbs light → bleached → opsin + trans-retinal
Na ⁺ channels	open — Na ⁺ flows in	opsin causes Na⁺ channels to close
Na ⁺ /K ⁺ pump	pumping out	still pumping Na⁺ out — "no change"
Rod membrane	depolarised	hyperpolarised
Glutamate (inhibitory)	released onto bipolar neurone	not released
Bipolar neurone	hyperpolarised — no impulse	depolarised → action potential in the optic nerve

MCQs: Oct 2019 Q6(a)(ii): light bleaches rhodopsin to **opsin and trans-retinal**; (iii): Na⁺ permeability **decreases**, pump activity **no change**. Oct 2021 Q3(a): chemical formed = **trans-retinal**; rod **hyperpolarised**, bipolar **depolarised**. Oct 2023 Q2(b)(i): photoreceptor = **opsin and retinal**. Oct 2025 Q5(a): pigment **rhodopsin**; channels **closed**, membrane **hyperpolarised**; ATP is used for **active transport of Na⁺ / reforming rhodopsin**.

35.3 The rhodopsin chain (asked five times)

Jan 2022 Q5(a)(ii) (3), June 2021 Q7(a)(ii) (4), Oct 2022 Q8(a) (4), Jan 2026 Q7(a) (3), Oct 2025 Q5(a):

rhodopsin absorbs light · splits into opsin and retinal / is bleached · cis-retinal → trans-retinal · (opsin causes) sodium ion channels to close — "*do not accept (opsin) blocks Na⁺ channel*" (Oct 2022) but "*(opsin) causes sodium ion channels to be blocked / closed*" (Jan 2026) — safest: "**Na⁺ channels close**" · more Na⁺ pumped out than diffuses in · membrane hyperpolarised · no glutamate released

Note: the June 2021 scheme's original wording "sodium ions are taken up" was **corrected before marking** to "*sodium ions are not taken up*".

35.4 Night vision and rods vs cones

- **Tarsier night vision (June 2026 Q2(b)(ii))**: rods are **sensitive to dim light** · **several rods synapse with one bipolar neurone (retinal convergence)** · **small generator potentials summate** · **triggering action potentials in the bipolar neurone** — "*to get mp4 they need to refer to bipolar neurone*". Large eyes → **larger pupil / retina** → **more light** to find prey.
- **Night monkey vs squirrel monkey (June 2024 Q3(a))**: rods detect **low light**, cones need **high light** for colour · the **day-active** squirrel monkey has **more cones** · the **night** monkey has **more rods** · both need colour vision, so cone numbers are of the **same order of magnitude**.
Calculation: rods = $3.5 \times 10^7 \div (3.5 \times 10^7 + 7.0 \times 10^6) = 5/6 \rightarrow 5000 \times 5/6 = 4167$.
- **RPE65 blindness (Jan 2021 Q5(a))**: active site **no longer complementary to trans-retinal** · **cis-retinal cannot be regenerated** · **rhodopsin cannot re-form** · rods have **no photopigment**.
Gene therapy (5(b)): higher dose → **more cis-retinal** · better **path-following** · reference to **error bars**.

35.5 The pupil reflex

1. **Light is detected by photoreceptors (rods / cones) in the retina.**
2. **Impulses** pass along **sensory neurones (optic nerve)** to the brain (a **reflex arc**: sensory → relay → motor).
3. **Motor neurones** carry impulses to the **iris muscles**.
4. **Bright light: circular muscles contract, radial muscles relax** → **pupil constricts**, less light enters (parasympathetic).
5. **Dim light / fight or flight: radial muscles contract, circular muscles relax** → **pupil dilates** (sympathetic; adrenaline).

Circular and radial muscles are **antagonistic**. Asked in Oct 2020 Q5(a)(iii) (4), June 2021 Q7(a)(i) (3), June 2023 Q1(b)(ii) (3), Oct 2023 Q2(a) (4), Oct 2025 Q5(b)(i) (2) — "**DO NOT ACCEPT ciliary muscles**" — and MCQ Oct 2021 Q3(c): increasing light → **circular contract, radial relax**.

- **Phenylephrine (Oct 2019 Q6(b))**: greater dose → **greater, longer-lasting, faster dilation** · it stimulates **Ca²⁺ release in the radial muscle** → **radial muscle contracts**, circular relaxes.
- **Pupil and learning (Oct 2025 Q5(b))**: children's pupils are **larger**; tasks **dilate** the pupil; children's dilation **peaks at difficulty 5**, adults' still rising at 8. Children's brains are **less developed / fewer synapses in the cerebral hemispheres**.

Pupil calculations:

- % increase bright → dark: $(6.41 - 3.35) \div 3.35 \times 100 = \mathbf{91.34\%}$ (June 2023 Q1(a)).
- Light passing the pupil: $r = 8\text{ mm} \rightarrow \pi r^2 = 201\text{ mm}^2 = 0.000201\text{ m}^2 \times 1400 = \mathbf{0.28}$ (Oct 2020 Q5(a)(ii)) — the m² conversion is a mark.
- From area 38.75 mm²: $d = 2\sqrt{(38.75 \div \pi)} = 7.02\text{ mm}$; increase from 4 mm = **3.02 mm** (Oct 2023 Q2(b)(iii)).
- Children's area: $r = 4.92 \div 2 = 2.46 \rightarrow 19.0\text{ mm}^2$; $- 16.9 = \mathbf{2.1\text{ mm}^2}$ (Oct 2025 Q5(b)(ii)).

36. Habituation and learning

Habituation (Jan 2026 Q5(a)(i)): a **decreased response to a repeated / prolonged / non-threatening stimulus** — *"ACCEPT where an organism stops responding"*.

Why it is an advantage (June 2021 Q5(a)(ii) MCQ): *"it filters out what is important and what is not important to react to"* — saving energy for real threats.

Mechanism (Oct 2025 Q6(c)(i)): with repeated stimulation, **fewer Ca²⁺ channels open** in the presynaptic membrane → **less neurotransmitter** → **less depolarisation** of the motor neurone → smaller response. (*"ACCEPT less depolarisation occurs / calcium ion channels become less responsive."*)

Statement MCQ (Jan 2022 Q3(a)(i)): "anatomical adaptation" **X**; "only in mammals" **X**; "only after repeated exposure to the same stimulus" **✓** → **1**.

36.1 Designing a habituation investigation — the points credited

From Oct 2019 Q7(a) (*Drosophila* climbing assay) and the investigations in June 2021 Q5, Jan 2024 Q3(b), Oct 2025 Q6(c):

apply the **same stimulus with the same force** · at **regular time intervals** · **record the response** (e.g. number of flies climbing, body contraction, change of direction) · **habituation has occurred if the response decreases** with repetition · control variables (temperature, light, age) · **test recovery** after a rest

36.2 Data questions

- **Fiddler crabs (June 2021 Q5(b))**: males and females **respond differently** · females **habituate more** · **variability after 10 exposures**. Test for the effect of number of encounters: **correlation coefficient** (not a t-test).
 - **Roundworms (Jan 2024 Q3(b))**: % difference 30 min after habituation = $(20 - 12) \div 16 \times 100 = 50 \%$ · all groups **start to recover after 0.5 min** · recovery **faster up to 10 min** · **the shorter the tapping interval, the faster the recovery** · some worms **have not recovered after 30 min**.
 - **Children and food pictures (Oct 2025 Q6(c))**: clicks fall because children **stop responding** — they **learn no food / reward comes** · % difference = $70 \div 165 = 42 \%$ (2 s.f.) · high-BMI children click **more** and their **decrease is smaller** · the difference is **significant after the sixth period** (error bars no longer overlap) · methodology (how "high BMI" defined, recent meals).
 - **Rats in a maze (Jan 2026 Q5(b))**: ratio day 4 = $2 : 2.4 : 5 = 1 : 1.2 : 2.5$ · group 1 (rewarded daily) **mistakes fall fastest** — they learn to reach the food · group 3 **fluctuating / slow fall until day 6**, then a **steep fall once food is present**.
 - **Elk habituated to cars (Jan 2026 Q5(a)(ii))**: two dangers — elk **hit by cars**, people **hurt in crashes / bitten**, feeding harms elk, **zoonoses**.
 - **Critical window (June 2019 Q6(a)(i))**: a **period of time during which a stimulus is required for normal development** — here, exposure to low O₂ for **development of the ventilation centre**.
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PART E — PLANT RESPONSES

37. Phytochrome

37.1 The two forms

	Pr (P660)	Pfr (P730)
Absorbs	red light, 660 nm	far-red light, 730 nm
Converted to	Pfr (rapidly)	Pr (rapidly)
In the dark	stable	slowly converts back to Pr / breaks down
Activity	inactive	physiologically active — promotes germination, affects flowering

- Sunlight contains **more red than far-red**, so during the day **Pr** → **Pfr** and Pfr accumulates; overnight it slowly reverts.
- **MCQs**: photoreceptor pigment in plants = **phytochrome** (*June 2019, Oct 2019, June 2025*); *Oct 2022 Q6(a)*: absorbs far-red **Pfr**, breaks down in dark **Pfr**; *June 2026 Q7(a)(i)*: breaks down in dark **Pfr**, absorbs red **Pr**.
- *June 2019 Q4(b)*: difference in maximum absorbance = **0.26 a.u.** (not nm — 70 nm is the gap between the peaks).

37.2 How active phytochrome promotes germination (June 2026 Q7(a)(iii), 3 marks)

Pfr enters the nucleus · binds to / activates a transcription factor (e.g. PIF) · transcription of genes involved in germination · consequence: gibberellin produced → amylase → starch broken down, cell division

The single most important rejection in Part E: "DO NOT ACCEPT Pfr acts as a transcription factor" (*Jan 2026 Q7(b)*) and **"IGNORE Pfr acts as a transcription factor"** (*June 2026 Q7(a)(iii)*). Pfr **activates** transcription factors; it is not one.

37.3 Germination experiments — the rule and the explanation

Rule: the last flash wins. If the last light treatment is **red**, Pfr is high and seeds **germinate**; if it is **far-red**, Pfr is converted back to Pr and they **do not**.

- *June 2026 Q7(a)(ii)*: dark 4 % · R 88 % · R FR 2 % · R FR R 98 % · R FR R FR 4 % → the only valid conclusion is **"Pfr causes germination"** (the **first** treatment does not decide it; Pr does not cause it).

- Oct 2025 Q2(a): none / R then FR / R alone. Group B germination 48 % of 250 = **120**. Chain: **red converts Pr → Pfr** · Pfr activates genes / transcription for germination · **red alone (group C) gives the highest germination** · far-red converts Pfr back to Pr · in the dark Pfr slowly reverts · less gene activation → fewer seeds germinate.
- Lettuce seeds (June 2019 Q4(c)): far-red inhibits germination · more red light → more germination · red at **48 h needs less intensity** than at 3 h (% difference **69 %** or **221 %**, depending on the base) · explanation: **red converts Pr → Pfr, more intensity → more Pfr**, in dark / far-red Pfr → Pr, Pfr stimulates germination, Pfr made **too early (3 h) reverts to Pr** before the seed can respond.

37.4 The phytochrome levels questions

Plantain seeds (June 2021 Q7(b), 6-mark *) — levels stated explicitly: Level 1 = **2 specific results** from the table · Level 2 = **chemical nature** — Pr/Pfr, absorption peaks, Pfr active · Level 3 = **detailed sub-cellular mechanism**, e.g. **effect on transcription** (Pfr → transcription of genes → enzymes for **gibberellin** → germination).

Lettuce seeds (Jan 2026 Q7(b), 6-mark *) — the scheme marks three strands: **B** (basic knowledge: R → Pfr, FR → Pr, slow dark reversion, Pfr active), **E** (explaining each group), **A** (additional: Pfr activates **transcription, amylase gene, starch** → **maltose/glucose**, respiration for **cell division**). 6 marks = **two E explanations from two groups + two A explanations**.

Group	Treatment	Explanation (E)
1	red	red Pr → Pfr → Pfr causes germination
2	darkness	no Pfr / Pfr slowly → Pr → no germination
3	red, darkness	dark not long enough to revert Pfr → germinates
4	red, far-red, darkness	far-red Pfr → Pr → not enough Pfr → no germination
5	far-red, red, darkness	last flash red → Pfr → germinates

37.5 Flowering and photoperiod

- **Short-day plants** are really **long-night** plants: they flower when the **night exceeds the critical night length** — i.e. when **Pfr is low**. A **flash of red** in the night **prevents** flowering; a following **far-red** flash **cancels** it.
- **Long-day (short-night) plants** flower when **Pfr is high** — a **red flash** in a long night **allows** flowering.
- Oct 2019 Q1(a)(i): the plant needing **shorter day than dark** = **pea**. Variables to control when testing flowering (1(b)): **temperature, light intensity/wavelength, water/humidity, soil pH, mineral ions, age of plant** — *"IGNORE species of plant"*.

- **Phytochrome and gibberellin in flowering (Oct 2023 Q6(c), 6-mark *)**: credit max one point from each data section (short-day, long-day, gibberellin graph — flowers **peak at about $1.5 \mu\text{mol dm}^{-3}$** then decline), then up to two each on **gibberellin action** (transcription factors, gene expression) and **phytochrome** (Pfr/Pr interconversion, active Pfr activates / represses flowering genes, **night-length importance**). Level 3 = **one detailed discussion of the effects of light and gibberellin**.

38. Auxin (IAA)

38.1 Phototropism — the chain

1. Auxin is made in the **shoot tip**.
2. With **unilateral light**, auxin **moves to / accumulates on the shaded side**.
3. It **stimulates cell elongation** on the shaded side.
4. The shaded side grows faster, so the shoot **grows towards the light** — **positive phototropism** — *"do not accept bend towards light"* (Oct 2022 Q6(b)).

38.2 The mechanism — acid growth (June 2025 Q6(c), 4 marks)

auxin **binds a receptor** · **H⁺ pumped / released into the cell wall** (proton pumps) · **pH in the cell wall falls** · this **breaks the hydrogen bonds between cellulose microfibrils** (activating expansins) · **K⁺ uptake lowers the water potential** of the cell · **water enters by osmosis** · the cell **elongates**

38.3 The coleoptile experiments (Jan 2022 Q5(b), 6-mark *)

Experiment	Result	Shows
1: intact / tip removed / tip capped	only the intact coleoptile bends	the tip detects light and produces the substance
2a: plastic sheet under tip	no bending	the substance cannot pass through plastic
2b: gel under tip	bending	the substance diffuses through gel — it is chemical
3: tip replaced on one side, in dark	bends away from that side	the chemical diffuses down and causes curvature even in darkness

Levels: L1 **discuss some results**; L2 + **chemical nature** — auxin, phototropins, **auxin transported to the dark side**; L3 + **molecular mechanism** — **proton pumps**, **low pH activates enzymes**, **hydrogen cross-links in the wall disrupted**, **water uptake by osmosis**.

38.4 Other auxin questions

- **Apical buds removed (June 2024 Q3(b))**: auxin on the cut top → **growth towards light** and **lateral buds inhibited** (apical dominance); gibberellin → **increased stem length** and **lateral bud growth** — "ignore 'bending'".
 - **Auxin and Chlorella (Oct 2022 Q6(d), 4 marks)**: both auxins **increase cell number** vs control · **natural IAA more effective than synthetic NAA** · IAA best at **0.1 $\mu\text{mol dm}^{-3}$** · NAA best at **1 $\mu\text{mol dm}^{-3}$** · IAA more effective at **lower concentrations**.
 - **Roots in low light (June 2026 Q7(b)(ii), 6-mark *)**: less IAA in roots (7.5 ng g⁻¹ less) and **lower IAA flow in the phloem** at low light · **roots shorter and lighter, same number of side roots** · IAA made in **growing tips / meristems**, moves down → promotes **root cell elongation** at suitable concentrations · **low light** → **less photosynthesis** → **less glucose** → **less respiration / growth; less IAA synthesised** · error bars **do not overlap** → **significant**; methodology (species, number, light intensity not measured). % difference in root length: $16.8 \div 39.1 \times 100 = 43 \%$.
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39. Gibberellins

What a gibberellin is (MCQs): *"an organic compound that stimulates the action of transcription factors"* (Oct 2023 Q6(b)); its function is to **"regulate the transcription of genes"** (Oct 2024 Q2(a)(i)). It is **not** "an enzyme that converts starch to glucose" and does **not** "inhibit elongation of cells".

39.1 Germination — the chain (Oct 2022 Q6(c), 4 marks)

1. The seed absorbs water; the embryo releases **gibberellin**.
2. Gibberellin **binds a receptor** in aleurone cells / **acts via transcription factors**.
3. The **amylase gene is switched on** — transcription → **mRNA** → translation → **amylase**.
(*"activates gene that produces amylase = mps 2 and 3"; "ACCEPT cause the inactivation of (amylase) gene inhibitor protein / repressor molecule"*.)
4. **Amylase hydrolyses starch to maltose / glucose** — "ignore monomers", "ignore amylose".
5. Glucose is used in **respiration** → **ATP for growth and cell division** of the embryo.

Why amylase is required (Oct 2025 Q2(b), 3 marks): amylase **hydrolyses glycosidic bonds** in starch · releasing **maltose / glucose** — *"DO NOT ACCEPT incorrect monosaccharides and disaccharides"* · used in **respiration / to make ATP** · **ATP / amino acids needed for growth / cell division** — *"IGNORE produces energy", "IGNORE germination"*.

39.2 Stem growth and data

- **Cereal internodes (June 2022 Q3(b))**: more gibberellin → **longer internodes** (all) · increase **mostly in the first internode**. Mechanism (3): gibberellin **enters the nucleus / binds a receptor** · **acts as / activates a transcription factor** · effect, e.g. **protein synthesis / amylase production** · **cell elongation / cell division**.

- **Ashoka seeds (June 2025 Q6(d)):** % change 200 → 400 ppm = $(46 - 114) \div 114 = -60\%$ (a decrease) · germination **rises to 200 ppm then falls** · gibberellin **activates transcription factors / genes** — *"IGNORE gibberellin acts as a transcription factor"*.
 - **Peanut seedlings in coloured light (June 2025 Q6(b), 4 marks):** growth in **all colours and both intensities** · **greater at high intensity** · **lowest in white light** · **greatest in red** · **one calculated rate or percentage** (e.g. red at high intensity 0.42 cm day^{-1}) — *"IGNORE data just read off the graph."*
-

PART F — GENE EXPRESSION AND GENE TECHNOLOGY

40. Transcription factors and switching genes on and off

Transcription factors are proteins that bind to the promoter region of a gene (or form a transcription initiation complex) so that RNA polymerase can bind and transcription can start.

- **Genes switched off by transcription factors (Oct 2023 Q1(b), 3 marks):** TFs are proteins · they bind to sections of DNA / the promoter · they can cause methylation / deacetylation of histones — DNA wrapped more tightly · preventing RNA polymerase binding — "ignore bases / enhancer sequence"; "ignore histone modification unqualified".
- **Cytokine gene expression in activated T cells (Oct 2021 Q6(c)):** activation produces / activates transcription factors ("ACCEPT secondary messenger systems") · they bind the promoter of cytokine genes · cytokine genes transcribed / mRNA produced.
- **Different cytokines activate different cells (Oct 2021 Q6(d)):** cytokines have specific shapes · bind complementary receptors on specific cells · initiating transcription / protein synthesis in T killer or B cells.
- **The tomato L-DOPA experiment (June 2022 Q4(b)(i), 4 marks):** the control gene / GM process has no effect (1.2 mg) · MYB12 alone has little effect (1.5) · CYP76 gives a large increase (10.2) because it codes for the enzyme · MYB12 + CYP76 gives most (14.6) because the transcription factor increases production of CYP76 protein.
- **Toxicant free radicals modulating gene expression (Jan 2022 Q8(d)):** they could change the shape of a transcription factor · prevent it binding DNA / promoters · so RNA polymerase cannot bind.

41. Epigenetics

Epigenetic modification changes gene expression without changing the base sequence.

Modification	Effect on DNA	Effect on transcription
DNA methylation — methyl groups added to cytosine (CpG sites)	TFs / RNA polymerase cannot bind	gene silenced
Histone acetylation — acetyl groups added to histones	neutralises positive charges on lysine → DNA less tightly wrapped	transcription possible
Histone deacetylation / methylation	DNA more tightly wrapped (heterochromatin)	transcription reduced

- **Statement MCQ (Jan 2023 Q4(b)):** "changes the base sequence" ✗; histone modification ✓; altered phenotype ✓; DNA methylation ✓ → 3.
- **Histones** give **structural support to DNA** (Jan 2024 Q5(b)(i)); **epigenetic changes** regulate gene expression (Jan 2024 Q5(b)(ii)).
- **Acetylation** → **more expression** (Jan 2023 Q4(d), 2 marks): acetylation **changes ionic charges / structure** of histones · DNA becomes **loosely wrapped** · **transcription can occur / RNA polymerase can bind** — "ignore genes are exposed", "ignore increased gene expression" (that is the question, not the answer).
- **DNA organisation in bats** (Oct 2025 Q8(d), 3 marks — "answer must be in logical sequence"): DNA methylation / histone modification · DNA **more tightly wrapped** around histones · TF binding / transcription reduced — "ACCEPT RNA polymerase cannot bind".
- **Prader–Willi syndrome** (Jan 2024 Q5(c)(i), 3 marks): maternal gene has **methyl groups** added to cytosine / CpG · paternal copy has the **gene / promoter deleted** · RNA polymerase cannot bind · **no transcription / mRNA** · **no protein** synthesised.
- **X inactivation** (June 2025 Q8(h)): one X in females is **inactivated / silenced** · by a **named epigenetic change** — DNA methylation, histone modification, non-coding RNA coating the X (Barr body).
- **Temperature determining sex** (June 2025 Q8(f)): temperature acts as a **stimulus** · **activates genes** for sex-hormone enzymes (e.g. **aromatase**, converting testosterone to oestrogen) · hormones bind **receptors** · initiating **sexual differentiation** ("ACCEPT reference to epigenetic changes").
- **Fibroblasts** → **myofibroblasts** (Oct 2023 Q8(e), 4 marks): respond to **TGF- β / chemical stimulus** · **change in gene expression / genes switched on** · by **epigenetic modification** · **active genes transcribed** · **mRNA translated** · **proteins for differentiation** (e.g. collagen).

42. One gene, many proteins: alternative splicing

The chain (Oct 2021 Q6(e); June 2023 Q7(b)):

the gene is **transcribed into pre-mRNA** · a **spliceosome** removes **introns** and **exons are joined in different combinations** (some exons removed) — **post-transcriptional modification** · **different mRNAs are translated** · proteins with **different primary structures / amino acid sequences** — and so different 3D shapes

- "Do not accept reference to DNA splicing" (Oct 2021) — **splicing happens to pre-mRNA**.
- **Three genes** → **14 serotonin receptors** (June 2023 Q7(b), 4 marks): **differential gene activation** by chemicals / TFs · transcription to **pre-mRNA** · **alternative splicing** → **14 mRNAs** · translation · receptors with **different amino acid sequences**.
- **Haemagglutinin stem vs head** (Jan 2021 Q8(f)): **different regions of nucleic acid code for stem and head** · **one version codes the stem, many the head** (more mutations in head genes)

— or **same exons for the stem, different exons for the head** (splicing).

43. Stem cells and differentiation

Pluripotent stem cell (Oct 2020 Q5(b)(i), 2 marks): a **self-renewing / continuously dividing** cell · that can give rise to **many / most cell types** — "**IGNORE** can give rise to all"; "**ALLOW** all cell types apart from extra-embryonic tissue".

43.1 How a stem cell becomes specialised — the chain

a **stimulus / chemical / transcription factor** · causes **differential gene expression** — some genes **switched on**, others off (including by **epigenetic modification**) · the **active genes** are **transcribed and translated** · producing **proteins that determine the cell's structure and function**

Asked in *Jan 2023 Q8(d)* (neurones), *Jan 2024 Q8(b)* (satellite cells → muscle: "*translation resulting in the synthesis of named proteins ... actin, myosin, troponin*"), *June 2024 Q8(f)* (zebrafish embryo: **mitosis, differentiation, colour gene switched on by TFs in skin cells, transcription and translation, red pigment**), *Oct 2025 Q7(a)(ii)* (myosin forms).

43.2 iPSCs (induced pluripotent stem cells)

- **Growth factors produce a functioning tissue (Oct 2020 Q8(g)):** they **bind receptors** · **activate transcription factors** · the cells **proliferate** · and **differentiate**.
 - **Studying Parkinson's with iPSCs (Oct 2019 Q7(b)):** obtain iPSCs **from people with Parkinson's** · **differentiate into neurones** · use cultured neurones for **drug testing / studying the genes involved**.
 - **Why check no stem cells remain in a retinal graft (Oct 2020 Q5(b)(ii)):** stem cells could **keep dividing** → **tumour** · or **differentiate into other cell types**.
 - **Advantages of reprogrammed cells for toxicity testing (Jan 2022 Q8(g)):** they carry the **individual's genetic background** · responses to toxicants are **individual** · **no embryos destroyed** → **fewer ethical issues**.
 - **Why blood vessels are needed in lab-grown spinal cord (Oct 2020 Q8(f)):** **mass transport** to overcome **diffusion limits** · delivering **oxygen / nutrients** · for **respiration** by the tissue.
-

44. Recombinant DNA and genetic modification

44.1 The standard chain — worth 3–4 marks in ten papers

1. **Isolate the gene** — cut it out with a **restriction endonuclease**, or **make it from mRNA with reverse transcriptase (cDNA)**, or **synthesise it**.
2. **Cut the plasmid with the same restriction enzyme** → **complementary sticky ends**.
3. **Insert the gene into the plasmid (vector)** and **join the sticky ends with DNA ligase** (phosphodiester bonds) → **recombinant DNA**.
4. **Put the vector into the host cell** — bacteria: **heat shock / calcium chloride**; plants: **Agrobacterium tumefaciens, gene gun**; animal cells: **microinjection, liposomes, viruses, electroporation**.
5. **Select the transformed cells (marker genes, e.g. fluorescence)** and **grow them in a fermenter / culture**; **extract and purify** the protein.

Paper	Organism / product	Marks
Oct 2021 Q2(a)	rice with bacterial β -carotene genes (Golden Rice)	4
Jan 2022 Q7(c)	adenovirus carrying the AAT gene (gene therapy)	4
June 2022 Q4(b)(ii)	tomato with beetroot CYP76 gene	3
Oct 2022 Q7(a)(i)	sheep producing rHE in milk	4
Oct 2023 Q5(b)	<i>E. coli</i> with periwinkle serotonin enzymes	4
Jan 2024 Q5(c)(ii)	<i>E. coli</i> making human growth hormone	4
June 2024 Q8(c)	red fluorescent zebrafish	4
Jan 2025 Q5(c)(i)	bacteria making human insulin	3
June 2025 Q7(d)(i)	mouse cells making human GH	4
Jan 2026 Q9(g)	microorganisms modified for taste	3
June 2026 Q1(b)(i)	maize with bacterial CSP8 — role of enzymes	2
Jan 2021 Q8(e)(i)	insect cells making Ebola antigen	4

The rulings:

- *"gene isolated using restriction enzyme = 2 marks"* (Oct 2021, Oct 2022) — but *"Please do not give 2 marks for endonucleases cutting gene and plasmid"* (June 2026). Name **ligase** separately.
- *"IGNORE restricting enzyme"* (Jan 2025) — the term is **restriction** enzyme / endonuclease.

- "ignore gene gun" for **bacteria** (Oct 2023, Jan 2024) — use **heat shock**. "IGNORE gene being inserted into bacterium" — it is the **plasmid** that is inserted.
- "ignore vector" where the question needs **plasmid** (Oct 2023).
- "ACCEPT reverse transcriptase transcribes mRNA into cDNA"; "ACCEPT using DNA polymerase to make copies of the gene" (June 2026).
- Sheep: the vector goes into **udder / milk-producing cells, a fertilised egg or embryo** ("ignore bacteria").
- Zebrafish: **shine UV light, select the glowing fish for breeding** (June 2024).

44.2 DNA MCQs that come with GM questions

- Enzyme used to make recombinant DNA: **restriction endonuclease** (Oct 2023 Q1(a)(i)).
- DNA in a bacterial cell: **plasmid and cytoplasm** (Oct 2023) — bacteria have no nucleus.
- Bond joining nucleotides in one strand: **phosphodiester** (Oct 2023; June 2026).
- A DNA mononucleotide contains **deoxyribose and thymine** (June 2026).
- A GM organism: **an organism containing genetic material from another species** (Oct 2024 Q7(a)).
- tRNA **identifies an amino acid and transports it to a ribosome**; cytokines are modified in the **Golgi apparatus**; DNA polymerase **adds complementary nucleotides / forms phosphodiester bonds / proofreads** (Oct 2021 Q6).
- Peptide bond + water drawn between two amino acids, joined by **condensation** (Jan 2025 Q1(a)).

44.3 Why choose a eukaryotic host?

- **Insect cells for Ebola antigen (Jan 2021 Q8(e)(ii))**: safer than growing Ebola · insect cells are **eukaryotic** — have **ER and Golgi** · protein is **correctly glycosylated / post-translationally modified** · low mutation rate.
- **Mouse cells vs E. coli for GH (June 2025 Q7(d)(ii))**: advantage — **protein more similar / more active / less immune response / correctly folded**; disadvantage — **slower division, more expensive, lower yield, ethical issues, contamination with mammalian pathogens**. "time / cost / yield could be an advantage or disadvantage depending on their reasoning."
- **GM bacteria vs animal insulin (Jan 2025 Q5(c)(ii))**: **no slaughter of animals** · large quantities / cheaper · no ethical / religious objection · fewer allergic / immune reactions · no transfer of animal infections.
- **Antibody from GM rice (Oct 2024 Q7(b))**: all produce BVZ · all have **higher binding activity than non-transformed rice** · animal culture highest · **strain B best** of the GM rice. Antibody action (7(b)(ii)): **complementary to antigens on cancer cells** · binds · **enhanced phagocytosis / opsonisation / agglutination / marks for T killer cells**.

45. Risks and benefits of GM organisms

The bank of credited points (Oct 2021 Q2(b); Oct 2022 Q7(b) *; Oct 2024 Q7(c) *; June 2024 Q8(d); June 2026 Q1(b)(ii)):

Benefits	Risks
higher yields / less crop damage → food security	gene transfer to wild relatives / other species (pollen, cross-breeding, horizontal gene transfer) → superweeds / superpests
pest resistance (Bt) → less insecticide; herbicide tolerance	unknown long-term effects of eating GMOs / on food chains
nutritional improvement (β -carotene) → less malnutrition	allergies; accumulation of GM material in the environment / non-target organisms
drought resistance → arid land can be farmed	monoculture → reduced biodiversity
medicines from GM organisms (pharming)	cost of seed / food; companies control the technology
reduced toxin in food (GM fungus → safer maize)	ethical objections — "unnatural"

How to reduce a risk (Oct 2021 Q2(b)(ii)): grow in **secure greenhouses**, keep **GM and non-GM crops separate** (buffer zones), use **sterile varieties**.

Scheme rulings: for drought-resistant maize (June 2026), "*IGNORE uses less water*" and "*IGNORE monoculture / decrease in biodiversity / superweeds*" — that scheme wanted **yield / farmers' profit / food security, arid land usable, cross-breeding / horizontal gene transfer, unknown long-term effects, higher seed cost**. Read what the **specific GM trait** does and tailor the benefit to it.

The GM levels questions — structure:

- **GM crops in the USA (Oct 2022 Q7(b)):** L1 = **table or graph** commented on (1) / **both** (2); L2 = + basic discussion of **both** benefits and risks, or detailed discussion of **one**; L3 = **detailed discussion of both**, "selective, concise and logical". Data: GM soybean **7 %** → **95 %**, GM corn **6 %** → **88 %** of area, 1996–2017.
- **GM fungus on maize in a warming world (Oct 2024 Q7(c)):** plant height, grains per cob up; toxin down (SDs **do not overlap for toxin** → **significant**; **do overlap for height and grain number** → **not significant**) · infection predicted to rise **38 %** → **73 %** → **95 %** with +2 °C and +5 °C (**+92 %** and **+150 %**) · benefits (outcompetes normal fungus, **safer food, food security, less pesticide**) · risks (**gene transfer to other fungi, toxin in food chains, cost, ethics, unknown effects**).
- **Zebrafish (June 2024 Q8(d), max 2 each):** benefits — **medical research, study regeneration / cell movement, pets / income**; risks — **visible to predators, health of fish, gene escaping**

into wild populations / food chains, illegal pet trade.

46. Finding active genes: microarrays, sequencing and bioinformatics

46.1 The microarray route — asked in eight papers

extract mRNA from the cells · use **reverse transcriptase to make cDNA** · label it with a **fluorescent dye / tag** · **hybridise to a microarray** carrying probes for thousands of genes · spots that **fluoresce show which genes are active / being transcribed** · compare **many samples** (diseased vs healthy) · use **bioinformatics / computer algorithms** to analyse the huge data set

- *Jan 2024 Q5(a)*: **microarray** · **detects mRNA** · **cDNA / fluorescent labels**.
- *Oct 2020 Q7(c)* (Parkinson's): identifies **active genes** · **many genes in a single sample** · data from **many individuals with and without the disease** · **bioinformatics** · **key differences identified**.
- *Jan 2021 Q5(c)* (blindness genes): microarrays for genes **active in the eye** · **sequence genomes** of people with inherited blindness · **compare many individuals with and without blindness**.
- *Oct 2022 Q8(b)*: two routes, "*Do not pick and mix*" — **microarray route** (mRNA, microarray, fluorescent probe, bioinformatics) **or PCR route** (mRNA → **reverse transcriptase** → **cDNA** → **PCR** → **gel electrophoresis**, compare bands).
- *Oct 2021 Q8(f)*: showing bacterial genetic material is **active** = find its **mRNA** with a **microarray** + **bioinformatics**.
- *June 2023 Q8(g)*: "*go down both routes and take highest score*" — extract mRNA / cDNA / PCR / probe / microarray / bioinformatics.
- *Oct 2024 Q8(f)* (dementia genes): **sequence genomes / use DNA or mRNA of people with and without dementia** · **microarrays / probes / PCR** · **genes only found in people with dementia** · **family history**.
- *Oct 2025 Q8(h)* (bat lifespan): **sequence bat genomes** · **microarrays for active genes** · **bioinformatics** · **compare bats that die young with long-lived ones**.

46.2 Showing two genes are related (homology)

- **Human–zebrafish homology (Jan 2022 Q8(c))**: **sequence the genes / DNA profiling** · **compare the sequences** · **show the products have similar functions**.
- **Piezo 2 found from Piezo 1 (Oct 2022 Q8(f))**: **compare DNA base / amino acid sequences** · **Piezo 2 has a similar structure** · **both respond to mechanical pressure**.

- **Whitefly gene from a plant (June 2024 Q8(a)):** sequence the whitefly genome · bioinformatics · compare with plant genomes · the gene found only in plants, not other whiteflies — or PCR / restriction / electrophoresis route. *"please don't mix and match."*
- **Why similar amino acid sequences give similar functions (Oct 2019 Q7(c)):** primary structure determines **bonding**, hence **folding / 3D shape**, hence **similar active / binding sites**.

46.3 Mutations, cancer and CRISPR

- **Mutation (Jan 2023 Q4(a)(i)):** a change in the base sequence of DNA / a gene — *"do not accept genetic information"*. Tumour cells divide by **mitosis** — *"do not accept cell division"*.
 - **Gene variants (Oct 2020 Q8(a)):** alternative versions / alleles of a gene · due to **changes in base sequence**.
 - **A variant enzyme reducing a drug's effect (Oct 2020 Q8(e), 4 marks):** altered **base sequence** · altered **primary structure** · protein **folds differently** · **active site no longer fits** the drug.
 - **Free radicals damaging (mt)DNA (Jan 2022 Q8(e); Jan 2023 Q8(c)):** **phosphodiester bonds broken / hydrogen bonds broken** · **mutation / change in base sequence** · repair enzymes affected. ROS are **made in mitochondria by oxidative phosphorylation**.
 - **CRISPR (June 2025 Q8(c)):** **guide RNA complementary to the target DNA sequence** · guides **Cas9 (endonuclease)** to **cut the gene** — *"REJECT plasmid"; "IGNORE enzymes unqualified"*.
 - **Genetic adaptations (Oct 2025 Q8(a)):** **inherited alleles** · that give an **advantageous phenotype / selective advantage**.
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PART G — THE SCIENTIFIC ARTICLE QUESTION

47. How the 20-mark article question works

The article is **released before the exam** and a clean copy is **enclosed as an insert**. The final question (Q8, or Q9 in Jan 2026) is **20 marks — 22 % of the paper**, split into **6–10 parts of 1–5 marks**, each tied to a **numbered paragraph**.

47.1 The articles, series by series

Series	Article	What the questions tested
Jun 2019	<i>Venom</i> (Natural History Museum)	neurotoxins, K ⁺ channels, α -latrotoxin, protein solubility, antimicrobial test design, natural selection, fMRI, learning
Oct 2019	<i>First in Fly: Drosophila</i>	habituation assay, iPSCs, protein structure, lipase, diazepam, dominant/recessive, non-disjunction, drug development
Oct 2020	<i>Personalized medicine</i> (National Geographic)	gene variants, immunotherapy, CT, checkpoint inhibitors, enzyme shape, stem-cell tissue
Jan 2021	<i>Developing vaccines</i> (Education in Chemistry)	Ebola, antibody production, H-bonding, egg-grown virus mutation, GM insect cells, splicing
Jun 2021	<i>Skeletal muscle performance and ageing</i>	muscle fibre, myokines, fibre types, Ca ²⁺ –troponin, creatine phosphate, tendons, myostatin
Oct 2021	<i>Rethinking Alzheimer's</i> (New Scientist)	brain regions, fMRI vs CT, amyloid vs bacteria, inflammation, gingipains, microarrays
Jan 2022	<i>Assessment of food toxicology</i>	Parkinson's, microarrays, homology, free radicals, mtDNA, MTT assay, iPSCs
Jun 2022	<i>Autonomic / enteric nervous system; POTS</i>	breathing control, neurotransmitter receptors, ECG, peristalsis reflex, ischaemia, inflammation
Oct 2022	<i>How we perceive the world</i> (Nobel 2021)	rods, microarrays, capsaicin/TRPV1, ion channels, Piezo, thermoregulation
Jan 2023	<i>Brain ageing: a mitochondrial point of view</i>	ROS, oxidative phosphorylation, mtDNA, differentiation, proteome, autophagy
Jun 2023	<i>Microbiota–gut–brain axis</i>	gut microbes, vagus nerve, T killer cells, fMRI, diet, MS, LPS receptor gene

Series	Article	What the questions tested
Oct 2023	<i>Your Second Skin / fascia</i>	sensory organ, connective tissue, pain, inflammation, fibroblast differentiation, lymph
Jan 2024	<i>Muscle tissue changes with aging</i>	sarcopenia, stem cells, testosterone, negative feedback, insulin, CT, mitochondria, protein diet
Jun 2024	Whitefly plant gene; GloFish zebrafish	bioinformatics, natural selection, GM zebrafish, risks/benefits, cell growth, differentiation
Oct 2024	Alzheimer's (The Conversation)	hypertension, amyloid stickiness, MRI + PET, ethics, monoclonal antibodies, placebo, inflammation
Jan 2025	<i>Thirsty work</i> (New Scientist)	hydration, caffeine, sodium homeostasis, osmosis, nitrate and vasodilation, ECG, lactose intolerance
Jun 2025	Y-chromosome loss and heart failure	microscopy, ratio, CRISPR, Biobank assessment, antibodies, X inactivation, bone-marrow immunity
Oct 2025	Fruit bats: sugar and cancer	genetic adaptation, pancreas as organ, kidney, epigenetics, anticoagulants, evolution, interferons
Jan 2026	<i>The Microbial Food Revolution</i>	sustainability, trophic efficiency, fermentation, probiotics, free radicals, RNA, GM, citations
Jun 2026	<i>Beyond tired: fatigue</i>	cross-bridge cycle, negative feedback, fMRI/PET, oxygen and ETC, heart/breathing control, inflammation, L-DOPA

47.2 The five kinds of sub-question

- Unit 5 chain in a new context (≈50 % of the marks).** Example: "*Explain the role of oxygen in the release of more energy*" (June 2026) = the chemiosmosis chain; "*Describe how the vagus nerve is involved in the control of heart rate*" (June 2023) = the heart-rate chain.
- Synoptic Units 1–4 chain (≈30 %).** Immune response, inflammation, protein structure, natural selection, osmosis, enzymes, cell organelles. You must carry these from AS and Unit 4.
- Design a method** — MTT assay, antimicrobial comparison, habituation assay, ECG study, Biobank assessment, fMRI study.
- Definitions / meanings of a phrase** — "sensory organ", "mitochondrial proteome", "genetic adaptations", "ischaemia", "monoclonal antibody", "sarcopenia", "LD₅₀".
- Small calculations and "science in society" marks** — ratio of chromosome lengths, % of genes lost (**87 %**), why a **placebo** was used, why it may be **unethical** to tell someone their risk, **why references are numbered** (Jan 2026 Q9(h): "*to identify the sources of information · so the original work can be read / cited / verified*").

48. The synoptic chains the article question keeps using

These come from Units 1–4 but are **marked with Unit 5 schemes**. Learn the scheme's wording.

The immune response (Jan 2021 Q8(b), 4; Oct 2022 Q7(a)(iii), 4):

antigen is **engulfed by a macrophage / APC** · **presented to T helper cells** · activated T helper cells **release cytokines that activate B cells** · B cells **differentiate into plasma cells** that **release antibodies** · **B memory cells** form

Inflammation (Oct 2021 Q8(d); Oct 2023 Q8(d); June 2022 Q8(h); Oct 2024 Q8(i); June 2026 Q8(f)):

injury / infection · release of **histamine / cytokines** · **vasodilation** → **increased blood flow** · capillary walls become **more permeable / leaky** · **fluid (oedema), white blood cells, antibodies** reach the tissue · **fever** reduces pathogen reproduction

"IGNORE blood vessels / arterioles / membrane" for the permeability point — say **capillary walls** (June 2026).

Protein structure and solubility / aggregation:

- June 2019 Q7(c): primary structure = **sequence of amino acids** · determines **folding / 3D shape** · **hydrophilic R groups on the outside** → water-soluble.
- Oct 2019 Q7(f): **misfolding** → **hydrophobic groups on the outside** → **bonds form between proteins** → **insoluble** aggregates.
- Oct 2024 Q8(b): β -amyloid is **fat soluble** · **hydrophobic / non-polar R groups** · **bonds / hydrophobic interactions form between protein molecules** → clumping — *"reject ... tails / fatty acids"*.

Natural selection (June 2019 Q7(f); June 2024 Q8(b); Oct 2025 Q8(f)):

random mutation → **new allele** · a **selection pressure** (toxin, pathogen) · individuals with the allele **survive and reproduce** · **pass the allele to offspring** · **allele frequency increases** over generations

Other recurring synoptic items:

- **Lysosomes / autophagy (Jan 2023 Q8(h); Oct 2019 Q7(e)):** **vesicle forms round the damaged mitochondrion** · **fuses with a lysosome** · **enzymes digest it** — lipids broken by **lipase hydrolysing ester bonds** → **glycerol and fatty acids**.
- **Monoclonal antibody (Oct 2024 Q8(g)):** antibody **specific / complementary to one antigen** · made by **one clone of B cells / plasma cells / hybridoma**.
- **Interferons (Oct 2025 Q8(g)):** released by **virus-infected cells**, **bind neighbouring cells** · **stop the virus replicating / entering** uninfected cells.

- **Checkpoint inhibitors (Oct 2020 Q8(d)):** cancer cells make **checkpoint proteins** · inhibitors stop them **blocking immune-cell activation** · lymphocytes **activated to kill tumour cells**.
- **Tumours with many mutations and immunotherapy (Oct 2020 Q8(b)):** more mutations → more **altered proteins** · on the **membrane** · recognised as **antigens** · **immune response** · **T killer cells** destroy more cancer cells.
- **Virus mutation in eggs (Jan 2021 Q8(d)):** mutations during **replication** · **coat-protein amino acid sequence changes** · the **specific** immune response no longer matches.
- **Anticoagulants in vampire-bat saliva (Oct 2025 Q8(e)):** a **chemical / enzyme inhibitor in saliva** · that **blocks a named clotting step** (prothrombin → thrombin, fibrinogen → fibrin, platelets).
- **Gut microbes and polysaccharides (June 2023 Q8(a)):** humans **cannot digest** cellulose / pectin · microbes release **enzymes that hydrolyse glycosidic bonds** · releasing **monosaccharides** used in **respiration** — "*do not accept produce energy*".
- **Probiotics (Jan 2026 Q9(d)):** increase **gut flora** · after **depletion by antibiotics / disease** · **outcompete pathogens for space / nutrients** · **produce chemicals that inhibit pathogens** · reduce **inflammation**.
- **Plant vs animal food (Jan 2026 Q9(b)):** energy lost between **trophic levels** · animals must be **fed plants** · **methane** from livestock · plants **take in CO₂** · **less land** per unit energy.
- **Why microorganisms have lots of RNA (Jan 2026 Q9(f)):** they **reproduce rapidly** → **rapid protein synthesis** → **RNA needed for translation**.
- **Hypertension and dementia (Oct 2024 Q8(a)):** damages **brain capillaries / endothelium** · **narrowing / blockage / bleeding / atheroma** · **less oxygen** to brain cells. **Falling blood pressure (8(e)):** less blood flow → less **O₂ / glucose** → **less aerobic respiration / ATP** → **brain cells damaged / die**.
- **Nitrate and blood flow (Jan 2025 Q8(f)):** nitrate (→ **nitric oxide**) in the blood · **relaxes smooth muscle** of vessels · **vasodilation**.
- **Lactose intolerance (Jan 2025 Q8(h)):** reason **linked** to explanation — **cannot produce lactase** (gene missing / not expressed) → **cannot hydrolyse lactose**; or an **immune response to milk protein**.
- **Ethical questions:** placebo — "*to act as a control · to compare with the drug group · remove bias / psychological effect*" (Oct 2024 Q8(h)); telling people their risk — "*risk isn't certain · could be false positive · anxiety · insurance · right not to know*" (8(d)); objections to using mammalian retinas — "*harm / blindness to animals · anaesthetic harm · animals cannot consent*" — "*do not accept killing*" (Oct 2021 Q3(b)(iii)); justifying rats — "*useful animal model · similar physiology · not ethical to use human babies*" (June 2019 Q6(c)).

How to prepare the article before the exam

1. Read it three times. Number every paragraph in your own copy.
2. For **each paragraph**, write in the margin the **Unit 5 or Unit 1–4 topic it connects to** (e.g. "¶12 K⁺ channels → action potential").

3. For every **process** named, write the **mark-scheme chain** from this note next to it.
 4. Look up every **unfamiliar term**; expect a "state what is meant by" question on one of them.
 5. For every **experiment** described, ask: what is the independent variable, what is controlled, where are the error bars, what would the control be?
 6. For every **figure or table**, practise one calculation (ratio, % change, magnification).
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PART H — PRACTICAL AND DATA SKILLS

49. Statistics and error bars — the marks nearly every paper gives

49.1 The error-bar / SD rule, in the wording that scores

- **Error bars / SDs do not overlap** → the difference is (likely to be) significant.
- **Error bars / SDs overlap** → the difference is not significant.
- **Always name the groups:** *"don't accept {overlapping / not overlapping} SDs are (not significant / significant) unqualified"* (June 2023 Q3(b)(i)); *"ignore error bars overlapping without being specific as to which"* (Oct 2024 Q4(c)).

Model sentences from the schemes:

- *"no overlap in any of the standard deviations for any type of dementia and control, which suggests that the conclusions may be valid"* (June 2021 Q3(c))
- *"no significant difference (between the two ostriches) for muscle T as SD overlap; significant difference for muscle F / G as SD do not overlap"* (Oct 2023 Q7(b))
- *"the difference becomes significant after the sixth (two-minute time block) because the error bars no longer overlap"* (Oct 2025 Q6(c)(iii))

49.2 Validity and reliability comments

From the "comment on the validity" questions (June 2021 Q3(c); Oct 2023 Q7(b)(ii)) and the methodology points in almost every levels question:

- **small sample size** — e.g. only **n = 2** Zimbabwean Blue ostriches vs 34 South African Black
- **unequal group sizes**
- **other variables not controlled** — nutrition, activity, **age**, **sex**, BMI
- **only one study** — *"ignore no repeats"*
- **no error bars / SDs / statistical test** → cannot judge significance
- **correlation does not prove causation** (June 2023 Q3(b)(i); Jan 2023 Q7(a)(i))
- **small SDs** → less variability, **more repeatable** data

49.3 How error bars are produced (Jan 2025 Q5(d)(ii)–(iii))

Repeat each concentration (replicates) · calculate **means** · then **standard deviations / standard errors** — and error bars are used by **comparing their overlap**: *"IGNORE reliability / validity / repeatability"* as an answer on its own.

49.4 Choosing a statistical test

- Relationship between two variables (e.g. number of encounters vs % running home) → **correlation coefficient** (*June 2021 Q5(b)(ii)*).
- Difference between two means → **Student's t-test**.
- (Hardy–Weinberg and index of diversity are distractors from Unit 4.)

50. Designing investigations — what the schemes credit

50.1 The recurring method points

Situation	Credited points
Plant flowering / growth (Oct 2019 Q1(b))	control temperature, light intensity/wavelength, water/humidity, soil pH, mineral ions, age of plant
Exercise studies (Oct 2019 Q5(a)(ii); Oct 2024 Q3(c)(iii))	length and intensity of exercise, fitness / BMI, ambient temperature, food / drug / caffeine intake — and say how : e.g. <i>"same food intake for 24 h before — standardised meals", "same exercise bike / same course"</i>
Baseline measurement (Oct 2024 Q3(c)(ii))	measured before exercise to give baseline data / a starting point for comparison — <i>"ignore control"</i>
Further study to support a conclusion (June 2019 Q2(c)(ii))	larger group / longer period / include males · control height, mass, BMI · say how success is measured · compare the chosen value with other values
Antimicrobial comparison (June 2019 Q7(d))	bacterial lawn / broth · method of applying each substance · incubation detail (e.g. 24 h at 20 °C; <i>"IGNORE temperatures above 35 °C or below 10 °C"</i>) · test different microorganisms · different concentrations · compare diameter of clear zones · aseptic technique
Learning / avoidance (June 2019 Q7(j))	pair the sound with a sting · later play the sound · record avoidance · faster / stronger avoidance = learning
Isolated muscle fibres (Jan 2026 Q8(c))	solutions isotonic with cytoplasm · control = saline / water and ions · measure length before and after
Toxicity of two drugs (Jan 2022 Q8(f))	living cells · same concentrations · time to colour change / colorimeter
Assessing cardiovascular health (June 2025 Q8(d))	questionnaire · two named measurements (ECG, blood pressure, cholesterol, BMI, cardiac output, heart rate) · compare with national data / statistical test

50.2 Graph skills

- **Line of best fit** — "**DO NOT ACCEPT LINE OF BEST FIT THROUGH ORIGIN**" when the data don't go there (*June 2024 Q4(c)(i)*); scales must use **more than half the grid**.
 - **Gradient** of a best-fit line: 0.104 N per % (accept 0.08–0.14) (*June 2024 Q4(c)(ii)*).
 - **Extrapolate** with the line of best fit: serotonin at $17.5 \text{ mg kg}^{-1} = 400 \text{ mg cm}^{-3} = 0.4 \text{ g cm}^{-3}$ (*June 2023 Q7(c)(ii)*).
 - **"Describe the relationship"** = overall trend + a **plateau / change of gradient with the value where it happens** — e.g. bipolar p.d. **rises up to 12 a.u., then constant** (*Oct 2021 Q3(b)(i)*); pH **little change up to 420 s, falls after** (*Oct 2019 Q2(c)(i)*); cardiac output **linear up to 10 mph, then levels off / falls** (*Jan 2023 Q6(d)(i)*).
 - **"Describe" data with two variables** — heart rate and lactate both rise, **but lactate's relative rise is far greater** (22.5 % vs 308.5 %) (*Jan 2022 Q2(c)*).
-

51. Core practical links (examined further in Unit 6)

The papers keep returning to three practical ideas that Unit 6 also examines:

1. **Rate of respiration (respirometer):** CO₂ absorbed by **soda lime / KOH**, so the **fluid moves towards the organisms as O₂ is used**; control tube with **glass beads of equal volume**; **water bath** for temperature; volume = $\pi r^2 \times \text{distance moved}$. RQ needs a second run **without** CO₂ absorber.
 2. **Breathing and exercise (spirometer):** **tidal volume, breathing rate, minute ventilation = tidal volume × rate**; oxygen uptake from the **downward slope of the trace** (CO₂ absorbed by soda lime).
 3. **Habituation (e.g. snails / worms / flies):** repeat the **same stimulus at regular intervals**, record **time to re-emerge / response**, stop when **no response**, test **recovery**; control **temperature, stimulus strength, time interval**, and treat animals **ethically** (return to habitat, minimise stress).
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PART I — EXAM TECHNIQUE

52. The 6-mark levels-based (*) question

52.1 What it has been about

Series	* question	Context
Oct 2020	Q3(b), Q4(c)	ADH and MDMA brain swelling; heart and regular exercise
Jan 2021	Q4(b)	mitochondrial disease — locating ETC defects
Jun 2021	Q7(b)	red / far-red light and seed germination
Oct 2021	Q5(c)	aerobic and anaerobic respiration in crocodiles
Jan 2022	Q5(b)	coleoptile experiments — auxin
Jun 2022	Q6(b)(iii)	ageing and the heart's response to adrenaline
Oct 2022	Q7(b)	risks and benefits of GM crops
Jan 2023	Q7(c)	negative feedback and blood glucose
Jun 2023	Q6(b)	distribution and roles of Na^+ , K^+ , Cl^- , HCO_3^-
Oct 2023	Q6(c)	phytochrome, gibberellin and flowering
Jan 2024	Q6(b)	athletes vs non-athletes
Jun 2024	Q6(a)(ii)	lactate and stroke volume, athlete vs non-athlete
Oct 2024	Q7(c)	GM fungus, maize and global warming
Jan 2025	Q6(c)	exercise and death from CVD
Jun 2025	Q4(c)	bacteria, inflammation and lifestyle in ageing
Oct 2025	Q4(b)	aerobic and anaerobic respiration in diving seals
Jan 2026	Q7(b)	phytochrome and lettuce germination
Jun 2026	Q7(b)(ii)	IAA and root growth in low light

Pattern: exercise / respiration physiology — 7 of 18; plant responses — 5 of 18; GM risk–benefit — 2; homeostasis — 3. Every one gives **data (a graph, table or both)** and says **"use the information ... and your own knowledge"**.

52.2 The three generic grids

Level	Marks	"Explain" grid	"Comment / assess" grid	"Discuss" grid
1	1–2	explanation attempted, limited interpretation, mainly one piece of information	limited number of factors synthesised; no judgement	isolated elements , generalised comments, vague statements
2	3–4	occasional analysis of both pieces of information ; some linkages	some factors synthesised; straightforward but accurate judgement	adequate knowledge; consequences occasionally supported
3	5–6	sustained application to both pieces of information; well-developed, logically structured	most factors synthesised; detailed and accurate judgement	comprehensive knowledge; sustained linkage; logically structured

Note "**both pieces of scientific information**": if the question gives a graph **and** a table, an answer that uses only one is **capped at Level 1–2**.

52.3 The specific rules the schemes add

- **Description** → **chemistry** → **mechanism**. "Level 1: description of (a minimum of 2 specific) results ... Level 2: reference to specific chemical nature ... Level 3: detailed mechanism ... at a sub-cellular level eg effect on transcription" (June 2021 Q7(b); Oct 2021 Q5(c); Jan 2022 Q5(b)).
- **The switching-off statement**. "To get 6 there has to be one explanation using negative feedback" (Jan 2023 Q7(c)).
- **Counted comments**. Oct 2025 Q4(b): 3 marks = 3 comments, **one an explanation**; 5 = 5 comments incl. **two explanations + one general comment**; 6 = **3 explanations + 1 general comment** or 2 + 2.
- **Aspects**. June 2024 Q6(a)(ii): 1 = describe one aspect; 2 = both aspects; 3–4 = both + explanation; 5–6 = **why stroke volume differs AND why lactate differs**.
- **Two linked aspects + method**. June 2025 Q4(c): "6 marks = explanation of at least two aspects that includes links between them / methodology comments."
- **B / E / A strands**. Jan 2026 Q7(b): **two explanations from two groups + two additional mechanism** points for 6.
- **One ion only caps you**. June 2023 Q6(b): "If only one ion max 3" (Level 2), "max 5" (Level 3).
- **All data sources**. June 2022 Q6(b)(iii): Level 3 needs **all three graphs**, the **older-people effect** and **significance**.

52.4 The winning structure

1. DESCRIBE a trend in source 1, with a figure → EXPLAIN it (physiology)
2. DESCRIBE a trend in source 2, with a figure → EXPLAIN it
3. Go one level down: the cellular / molecular mechanism (ETC, transcription, Pfr, H⁺ pumps)
4. Link the two sources together
5. EVALUATE: error bars / SD – which groups, significant or not; sample size, age, sex

Six correct but unlinked descriptions score **Level 1–2**. The explanation and the mechanism are what move the answer up.

53. Command words

Command	What the mark scheme wants
State / Give / Name	a short fact — no explanation
Describe	what happens / what the data show, with figures ; no reasons
Explain	reasons — <i>because / so / therefore</i>
Suggest	apply biology to an unfamiliar context; several answers accepted
Calculate	working, unit , the requested s.f. / d.p. / standard form
Deduce / Determine	a conclusion from the data , with the reasoning
Compare and contrast	a similarity AND a difference , each written as a direct comparison — " <i>do not piece together</i> " (June 2022 Q5(a)(ii)); " <i>2 marks max for similarities, 2 marks for differences</i> " (Oct 2021 Q8(b))
Comment on	a judgement supported by the data — trend, figure, error bars / significance, methodology
Assess / Discuss	both sides / several factors, then a judgement
Describe conclusions / Describe three conclusions	statements the data support — each a separate, specific conclusion (e.g. June 2025 Q5(a)(i): "must refer to exercise and rest periods")

54. Calculations that recur

% change	= (new – original) ÷ original × 100	state INCREASE or DECREASE
% difference (when the question gives this)	= (difference ÷ mean of the two values) × 100	
ratio (keep the order asked)	= divide both by the smaller → x : 1	
magnification → nm (×1000)	= image size ÷ actual size	convert mm → μm (×1000)
actual size	= image size ÷ magnification	
area of circle	= πr^2 (radius = diameter ÷ 2)	
volume of sphere	= $\frac{4}{3} \pi r^3$	
gradient / rate	= $\Delta y \div \Delta x$, with units (e.g. a.u. ms ⁻¹ , pmol dm ⁻³ h ⁻¹)	
cardiac output	= heart rate × stroke volume	
cardiac index	= cardiac output ÷ body surface area (dm ³ min ⁻¹ m ⁻²)	
minute ventilation	= tidal volume × breathing rate	
RQ	= volume CO ₂ produced ÷ volume O ₂ used	
efficiency	= (ATP × energy per ATP) ÷ energy in substrate × 100	
heart rate (ECG)	= 60 ÷ time for one cycle (s)	
standard deviation	= $\sqrt{[\Sigma(x - \bar{x})^2 \div (n - 1)]}$	
antilog	= 10 ^x (read log values first, antilog, THEN subtract)	

Precision is marked — these are the rulings:

- "1.30 = 1 mark only" when 2 s.f. was asked (June 2023 Q3(b)(ii)); "69.44 / 69.4 gains 1 mark" (June 2025 Q5(b)).
- "If answer given to one decimal place e.g. 14.7 = 0 marks" when 2 d.p. was asked (June 2026 Q4(a)(i)).
- **Standard form** is a separate mark: "if not in standard form 15000–16000 gains 1 mark" (June 2023 Q4(b)(ii)).
- **Units** are a separate mark: "max 2 marks if no units / or incorrect units" (Jan 2022 Q7(a)); "no units max 1 mark" (Jan 2024 Q7(c)(i)); "IF THEY USE UNITS eg nm NO MP2" for a magnification (Oct 2024 Q6(a)(i)) — **magnification has no units**.
- **Measurements have tolerances** — the schemes give ranges (e.g. 44 ± 1 mm). Measure carefully with a ruler, in **mm**.
- **Correct answer with no working usually gets full marks — but a wrong answer with no working gets nothing**. Always show working.

Worked magnifications from the schemes:

Paper	Measurement → working	Answer
June 2022 Q7(a)	98 mm ÷ 0.00075 mm	1.3 × 10⁵
June 2023 Q4(b)(ii)	scale bar →	1.55 × 10⁴
Jan 2024 Q2(b)(i)	9.5 mm = 9500 μm ÷ 50 μm	1.9 × 10² (light microscope)

Paper	Measurement → working	Answer
Oct 2024 Q6(a)(i)	$52 \text{ mm} \div 0.00007 \text{ mm}$	7.43×10^5
June 2025 Q2(a)(iii)	$44 \text{ mm} = 44\,000 \mu\text{m} \div 2.7 \mu\text{m}$	1.63×10^4
Jan 2022 Q1(a)	width $24.5 \div$ length $75 \times 2000 \text{ nm}$	653 nm
Oct 2022 Q3(b)(ii)	$11\,000 \mu\text{m} \div 3300$	3.3 μm (SEM)
Jan 2025 Q7(a)(ii)	$30\,000 \mu\text{m} \div 23\,500$	1.28 μm (TEM)

Microscope names: a surface view of a nerve = **scanning electron microscope** — "*Do not accept transmission electron microscope*" (Oct 2022); a section through a mitochondrion = **electron microscope** — "*REJECT SEM*" (Jan 2025); $\times 190$ photograph of nerve cross-section = **light microscope** (Jan 2024). "*Do not accept electric / electrical / electronic*" — every time.

55. Mark-scheme phrase bank

Respiration

- "*electrons pass along the electron transport chain, releasing energy*" — never "producing"
- "*energy used to pump protons into the intermembrane space*"
- "*protons diffuse through ATP synthase*" → "*ADP + Pi → ATP*"
- "*oxygen is the final electron acceptor*" / "*forms water*"
- "*pyruvate is reduced to lactate, regenerating NAD so glycolysis continues*"
- "*lactate is converted to pyruvate in the liver*"
- "*RQ = volume of CO₂ produced ÷ volume of O₂ consumed*"

Muscles and the heart

- "*Ca²⁺ bind to troponin, troponin changes shape, tropomyosin moves, myosin binding sites exposed*"
- "*ATP binds to the myosin head, breaking the cross-bridge; ATP is hydrolysed, returning the head to its original position*"
- "*cardiac muscle is myogenic*"; "*SAN produces waves of depolarisation / excitation*"; "*AVN delays*"; "*bundle of His → Purkyne fibres*"
- "*chemoreceptors detect increased CO₂ / decreased pH*" → "*impulses to the cardiovascular control centre in the medulla oblongata*" → "*sympathetic neurones to the SAN*" → "*SAN depolarises more frequently*"

Homeostasis and kidney

- "thermoreceptors → thermoregulatory centre in the hypothalamus → impulses to named effectors"
- "afferent arteriole wider than efferent → high pressure → forced through basement membrane / fenestrations / podocytes"
- " Na^+ actively transported out of PCT cells into the blood; glucose enters by co-transport with Na^+ , then facilitated diffusion into the blood"
- "ascending limb actively transports Na^+ / Cl^- out, is impermeable to water; water leaves the descending limb by osmosis"
- "osmoreceptors in the hypothalamus → ADH from the posterior pituitary → aquaporins inserted into collecting duct membranes → more water reabsorbed by osmosis"
- "steroid hormone diffuses through the membrane, forms a hormone–receptor complex which acts as a transcription factor"
- "peptide hormone binds a membrane receptor, activates a second messenger (cAMP)"

Nerves

- "sodium–potassium pump actively transports 3 Na^+ out, 2 K^+ in; membrane more permeable to K^+ "
- "voltage-gated Na^+ channels open, Na^+ diffuse in, depolarisation, threshold"
- "myelin is an insulator; action potentials only at nodes of Ranvier; saltatory conduction"
- "voltage-gated Ca^{2+} channels open → vesicles fuse → exocytosis → neurotransmitter diffuses and binds receptors on the postsynaptic membrane → Na^+ channels open"
- "L-DOPA crosses the blood–brain barrier and is converted to dopamine in the brain"
- "rhodopsin → opsin + trans-retinal → Na^+ channels close → rod hyperpolarised"
- "circular muscles contract, radial muscles relax → pupil constricts"

Plants and genes

- "red light converts Pr → Pfr; Pfr activates transcription factors" (never "Pfr is a TF")
- "auxin → H^+ pumped into cell wall → low pH → hydrogen bonds between cellulose microfibrils broken → water enters by osmosis → cell elongation"
- "gibberellin → amylase gene transcribed → amylase hydrolyses starch to maltose / glucose → respiration"
- "restriction endonuclease cuts gene and plasmid → complementary sticky ends → DNA ligase → vector into host by heat shock / gene gun / microinjection"
- "extract mRNA → reverse transcriptase → cDNA → fluorescent label → microarray → bioinformatics"
- "differential gene expression — active genes transcribed and translated into proteins that determine structure and function"

56. The twenty-five most commonly lost marks — each one a mark-scheme ruling

1. **"Signals" or "messages"** for nerve impulses — *do not accept / REJECT*, in almost every scheme.
2. **"Produces energy"** — *REJECT*. Energy is **released**; ATP is **produced**.
3. **"Sodium"** without "ions" — *"do not accept sodium unqualified"*. Write **sodium ions / Na⁺**.
4. **"ATPase"** for the enzyme in chemiosmosis — *REJECT* (June 2026). Write **ATP synthase**.
5. **"ADP + Pi → ATP"** with no enzyme named — *IGNORE*. Name **ATP synthase**.
6. Saying ATP hydrolysis drives **the power stroke** — *do not accept*. ATP hydrolysis **re-cocks** the head.
7. **"Impulses" from the SAN** — *IGNORE*. It sends **waves of depolarisation / excitation**.
8. **"Control centre" unqualified** — *IGNORE*. Name the **thermoregulatory centre in the hypothalamus** or the **cardiovascular control centre in the medulla**.
9. **"Coordination" unqualified** for the cerebellum — *ignore*. Write **coordination of voluntary movement / balance / posture**.
10. **ADH "produced by the pituitary"** — the MCQ answer is **hypothalamus** (made there, released from the posterior pituitary).
11. **"Permeability of the tubule changes"** without naming it — *ignore*. Name the **DCT / collecting duct**.
12. **"More urine produced"** — *ignore*. Say **more dilute urine / more water excreted**.
13. **"Pfr acts as a transcription factor"** — *DO NOT ACCEPT / IGNORE*. Pfr **activates** transcription factors.
14. **"Bends towards light"** — *do not accept*. The shoot **grows towards the light** by cell elongation.
15. **"DNA splicing"** — *do not accept*. **Pre-mRNA** is spliced.
16. **Lactate "used in the link reaction / Krebs cycle"** — *ignore*. It is **converted to pyruvate** first.
17. **"Steroid hormones bind to cell-membrane receptors"** — *reject*. They bind **inside the cell**.
18. **"Genes are exposed" / "increased gene expression"** as the effect of acetylation — *ignore*. Say **DNA less tightly wrapped, so RNA polymerase can bind / transcription can occur**.
19. **Error bars "overlap" with no groups named** — *not accepted*. Name the groups and state significance.
20. **Comparing ECGs without naming P, QRS or T** — *"unless refer to labels PQRST they will not get the marks"*.
21. **Compare-and-contrast answers written as two separate paragraphs** — *"do not piece together"*.
22. **Wrong precision**: 1.30 for 2 s.f., 14.7 for 2 d.p., missing standard form, missing units.
23. **"Electric / electronic microscope"** — *REJECT*. And choose **SEM vs TEM** from the image type.

24. **"Cell division"** for how a tumour grows — *do not accept*. Write **mitosis**. **"Genetic information"** for a mutation — *do not accept*. Write **change in the base sequence of DNA**.
25. **Describing the data in a levels question and stopping** — capped at **Level 1**. Level 3 needs the **mechanism** and an **evaluation**.

57. Per-paper map

Paper	Question topics (* = levels question)
2019 Jun (WBI05)	myogenic, ECG heart rate, SAN · ACL tendon graft, keyhole surgery · negative feedback, hypothalamus in exercise, hormonal vs nervous · phytochromes and lettuce germination · ATP, NAD, alligator lactate*, oxygen debt, Krebs · critical window, ventilation centre · article: venom
2019 Oct (WBI05)	photoperiod and flowering · brain regions, fMRI, blood pH in exercise · ETC diagram, oxidative phosphorylation · joint tissues, ACL and osteoarthritis · blood flow in exercise, coordinated heartbeat* · rods, rhodopsin, phenylephrine and pupil · article: Drosophila
2020 Oct	anaerobic MCQs, mitochondrion drawing, chemiosmosis · nephron, ultrafiltration of urea, kangaroo rat · MDMA, ADH and brain swelling* · exercise heart rate, cardiac output, heart and exercise* · pupil and light, iPSC retina · myelin, resting potential, TTX · L-DOPA, dopamine release, microarrays · article: personalised medicine
2021 Jan	action potential MCQs, resting potential and ATP · training and fibre types · brain regions, CT/PET, seizure temperature · sarcomere MCQs, mitochondrial disease ETC* · RPE65 blindness and gene therapy · recovery after exercise, straight-line calc, heart-rate control · synapse, L-DOPA, TFs and dopamine, LRRK2 · article: vaccines
2021 Jun	respiration products, lipid equation, ATP efficiency, oxygen's role · reflex arc, Pacinian corpuscle · myelin, polyneuropathy, myelin and dementia · ADH MCQ, PCT microvilli, renal vein area, glucose/urea, urine volume · habituation: leech, fiddler crab · SAN/AVN, exercise and cardiac cycle, ECGs · pupil reflex, rhodopsin, red/far-red germination* · article: muscle ageing
2021 Oct	Ca ²⁺ in contraction, marathon fibres, heat loss · Golden Rice and GM risks · rods and bipolar cells, light intensity data, ethics · feedback, GFR, diabetes surgery data · RQ, hydramethylnon, metabolic rate, crocodile respiration* · tRNA, Golgi, DNA polymerase, cytokine genes, CD45 splicing · stress drugs and heart rate · article: Alzheimer's
2022 Jan	mitochondrion size, oxidative phosphorylation, ATP use · joints, myoglobin, sprinters, heart rate and lactate · habituation, reflex arc, diving whale heart rate · Bowman's capsule, countercurrent, ADH · rhodopsin, coleoptiles* · action potential, myelin gradient, PET cocaine · alveolar ventilation, ECG, adenovirus gene therapy · article: food toxicology

Paper	Question topics (* = levels question)
2022 Jun	brain labelling and functions · Bowman's capsule drawing, filtration MCQs, glucose reabsorption, urea · ADH MCQs, gibberellin and internodes · Co-beneldopa, GM tomato L-DOPA · synaptic cleft, sensory vs motor, Guillain-Barré, Ca ²⁺ channel mutation · adrenaline, cardiac output and index, ageing and adrenaline* · mitochondrion magnification, pyruvate/ADP data, fibre mitochondria, ATP in contraction · article: autonomic nervous system
2022 Oct	inner membrane, lactate, chemiosmosis, labelled oxygen · giraffe joints, ATP in sliding filament · relay neurone, lidocaine, SEM, myelin · heart rate and lifespan, elephant O ₂ , cheetah · camel, water use, kangaroo rats · phytochrome, auxin, gibberellin and amylase, IAA vs NAA · GM sheep rHE, anti-rHE antibodies, GM crops* · article: perceiving the world
2023 Jan	elbow joint, ligament · fMRI, PET vs MRI, brain functions · action potential, reflex arc, synapse · mutation, epigenetics, oestrogen, histone acetylation · water formation, cyclist RQ, COPD · fibre types, sarcomere, cardiac output/index, heart-rate control, lactate · sucrose and dopamine, SD, adrenaline, blood glucose feedback* · article: brain ageing and mitochondria
2023 Jun	pupil size, adrenaline, iris · ETC, reduced NAD, CO and cyanide · lung conditions, ventilation vs body mass, minute ventilation, exercise ventilation · tendons, sarcomere magnification, fibre types, force–velocity · filtration MCQs, countercurrent, urine data · ions across membranes, ion distribution* · serotonin, receptor splicing, MDMA · article: gut–brain axis
2023 Oct	recombinant DNA MCQs, TFs switching off genes · pupil reflex, photoreceptor, drugs, pupil area · ADEM lesion volume, MRI dye, walking · heart rate and lifespan, ECG traces, giraffe cardiac output · synapse, GM E. coli serotonin, culture data · adrenaline, nicotine Daphnia, gibberellin, flowering* · energy use in contraction, sliding filament, ostrich muscles, hypothalamus · article: fascia
2024 Jan	respiration order, glucose equation, lipid RQ and efficiency · depolarisation, repolarisation, nerve magnification, myelin · C. elegans reflex and habituation · nociceptors, nicotine knockout mice · microarrays, histones, Prader–Willi, HGH in E. coli · sprinters, lactate, athletes vs non-athletes* · kangaroo rat temperature and water balance · article: muscle ageing
2024 Jun	ACL MRI and calculation, antagonistic muscles · brain labelling, dopamine, SSRIs · rods vs cones in monkeys, auxin/gibberellin on seedlings · sarcomere ratio, troponin, ATP and tension graph · BMAA and ALS, PET vs CT, neurone drawing, TTX · lactate MCQ, lactate and stroke volume*, cyanide · nephron MCQs, PCT cell, drinking water data, hyponatraemia · article: whitefly and zebrafish

Paper	Question topics (* = levels question)
2024 Oct	neurone direction, action potential MCQs, sensory vs motor · gibberellin, adrenaline, TBT and steroid hormones · Ca^{2+} and fatigue, calcium homeostasis, cyclists' lactate · Parkinson's and age, carbidopa, dopamine and Drosophila · pronghorn heart/ventilation, metabolic rate, temperature in exercise · Bowman's capsule magnification/volume, osmoreceptors, overhydration, glucose filtration · GM definition, BVZ antibody from rice, GM fungus* · article: Alzheimer's
2025 Jan	peptide bond, deamination · brain labelling and functions · afferent/efferent, ADH and urine · action potential, K^{+} permeability gradient, ion ratio, temperature, myelin data · homeostasis, insulin and exercise, GM insulin, insulin and cell division · dopamine, lidocaine and heart cells, exercise and CVD* · TEM mitochondrion, chemiosmosis, amygdala, Krebs products, mitochondria in developing neurones · article: hydration
2025 Jun	motor neurone, synapse · cristae, mitochondrion magnification, reduced NAD, oxygen · lidocaine, MDMA, TTX and breathing, LD_{50} · L-DOPA, medulla, ageing: bacteria and inflammation* · exercise and blood pressure, systole/diastole control, stroke volume · phytochrome, coloured light and growth, auxin mechanism, gibberellin and germination · pituitary, GH, GH and age, GM mouse cells · article: Y chromosome
2025 Oct	glycolysis, pyruvate in aerobic and anaerobic respiration · phytochrome and thale cress, amylase · kidney section, urine and habitat, kangaroo rat nephron · seal ECG, diving seals* · rhodopsin, pupil and learning, pupil area · Schwann cell, saltatory conduction, synapse, habituation in children · fast-twitch structure, myosin genes, collagen/elastin · article: fruit bats
2026 Jan	nicotine and cobra toxin, log scale, ED_{50} · RQ definition and oxygen, barley RQ · reflex arc, Ca^{2+} channels, positive feedback · nephron magnification, glucose filtration/reabsorption · habituation, elk, rats in a maze · cardiac output, blood distribution, ECGs · rhodopsin, lettuce phytochrome* · fibre types, capillaries, muscle fibres in ATP · article: microbial food
2026 Jun	DNA MCQs, enzymes in recombinant DNA, drought-resistant GM maize · tarsier eye volume, rod cell, night vision · nephron, glucose reabsorption, medulla : cortex, fennec fox · penguin SA:V, thermoregulation, AVT and aquaporins · extensor, sarcomere, stroke volume, sprinter fibres, RQ in a 1500 m race · conduction velocity data, myasthenia gravis · phytochrome MCQs and germination, IAA and roots* · article: fatigue

The architecture is fixed. Every current paper has a **respiration** question, a **heart / exercise** question, a **nerves or synapse** question, and nearly always a **kidney** question and a **muscle** question, plus the 20-mark article. Prepare those five and the article technique, and you have about **65–75 of the 90 marks** covered.

58. What an A* requires, and the final plan

The rule

A* requires grade A for the IAL overall (≥ 480 of 600 uniform marks) and $\geq 90\%$ of the uniform marks across the IA2 units combined (≥ 270 of 300).

The IA2 units are **4, 5 and 6** — 120 + 120 + 60 UMS. **The A* is decided entirely by A2 performance.** Unit 5 is **120 of the 300**; and Unit 6 is examined on the practical work behind Units 4 and 5 — so §5, §11.4, §17, §36, §49–§51 of this note are Unit 5 content **and** Unit 6 content.

Priority order

Priority	Content	Marks/paper
1	Chemiosmosis, lactate and RQ chains (§4, §6, §8)	6–14
2	Synapse, action potential, myelin chains and drugs (§27–§31)	6–20
3	Heart-rate control, cardiac output, ECG (§13–§16)	6–15
4	Ultrafiltration, reabsorption, countercurrent, ADH (§22–§25)	4–17
5	The article question: synoptic chains + article annotation (§47–§48)	20
6	Sliding filament in order + fibre types (§11–§12)	3–14
7	Recombinant DNA, microarrays, epigenetics, splicing (§40–§46)	3–21
8	Levels technique: describe → explain → mechanism → evaluate (§52)	6
9	Phytochrome, auxin, gibberellin mechanisms (§37–§39)	1–13
10	Error bars, validity, calculations with precision (§49, §54)	8–14

Five-step routine for every question

1. Circle the COMMAND WORD and the MARK ALLOCATION – how many points?
2. Underline the CONTEXT and every data source – Level 3 uses all of them.
3. Calculations: working → answer → UNIT → requested s.f./d.p./standard form.
4. Explanations: the chain from §55, in ORDER, with the exact terms.
5. Re-read: have I named the error bars' groups, given a figure, and added a methodology point where the question says "comment"?

Exam-day tactics

- **Pace by the mark allocation** — about a minute a mark, and **25 minutes kept for the article question**.
- **Bring your annotated knowledge of the article** into the room: you get a clean insert, but the thinking is already done.
- In every * question, **alternate description and explanation**, go down to the **molecular mechanism**, and finish with an **evaluation**.
- **Quote figures with units** whenever you describe data; add **one calculated value** where the scheme asks for a "calculation to support".
- Write **impulses**, **sodium ions**, **releases energy**, **ATP synthase** — every time.
- Never leave a "**suggest**" blank — it is designed to be answerable from general biology.
- Leave five minutes to check that every **explain** has a *because*, every **calculation** has a **unit and the right precision**, and every **comment** has an **error-bar or methodology point**.